International Journal of Transportation Science and Technology 12 (2023) 148–172

Contents lists available at ScienceDirect

International Journal of Transportation Science and Technology

journal homepage: www.elsevier.com/locate/ijtst

# An integrated approach for optimizing location-inventory and location-inventory-routing problem for perishable products

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ARTICLE INFO

*Article history:*
Received 25 November 2020
Received in revised form 19 December 2021
Accepted 6 February 2022
Available online 09 March 2022

*Keywords:*
Perishable products
Direct shipment
Location-Inventory problem
Location-Inventory-Routing problem
Optimization
Mixed linear programming model

ABSTRACT

The grocery market, especially the perishable product market is under amazing developments in China, accounting for half of overall food expenditure. However, the supply chain of perishable products has encountered serious problems, including significant freight losses, high inventory and transportation costs, etc. To solve these problems, the current paper proposes an integrated approach for simultaneously optimizing the decisions involved in the supply chain, typically including location, inventory, and routing strategy. A three-echelon supply chain including suppliers, DCs and retailers is structured in this paper, with the impacts of whether allowing direct shipment from supplier to retailer being considered. Two typical problems are investigated: location-inventory problem (LIP) and location-inventory-routing problem (LIRP). Mixed integer nonlinear programming model for LIP is proposed while Mixed integer linear programming models for LIRP is proposed, with the objective of minimizing the overall costs of supply chain, which are composed of infrastructure construction costs, transportation costs, inventory costs, and freight losses. CPLEX is used for LIP model solution and a two-stage heuristic algorithm based on Simulated Annealing is designed to solve the LIRP model. The proposed mathematical models and solution method are then applied to a real supply chain network for fresh milk. The optimized results suggest that considering the direct shipment from supplier to retailer is beneficial in terms of reducing the overall supply chain costs by 31%, with 43% reductions in transportation costs, 69.4% reductions in freight losses, 99.4% increases in inventory costs. With the developments of e-retailers for perishable products, allowing the direct shipment from supplier to retailer will bring more benefits in the future.
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## Introduction

The grocery market, especially the perishable product market is under amazing developments in China. According to Nielsen Report, consumption of perishable products in Asia accounts for 50% of overall food expenditure (China Consumer Trends, 2018). The market is particularly popular in China, with 1.79 trillion trading volume (in yuan) in 2017 and 6.7% growth rate annually since 2010. However, the supply chain of perishable products has encountered serious problems here, due to the facts that the infrastructure coping with cold chain logistics is still under developments and an integrated solution

Peer review under responsibility of Tongji University and Tongji University Press.
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https://doi.org/10.1016/j.ijtst.2022.02.002
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with efficient coordination among all the processes involved in the chain is lacking. One of the encountered problem is the significant attrition rate of the goods. Statistics show that rate of damaged cargo is about 30% incurred in the process of transportation and inventory for perishable products in China, compared with 5% of damage rate in developed countries. Another problem is the sales revenue of perishable product retailers is very limited. In particular, around 80% of e-retailers in the market are under deficit situations currently.

Considering the promising potential of the market and the huge gap between developed countries and China, increasing interests have arisen from both government and scholars. From the perspective of government, China Ministry of Finance and Ministry of Commerce have jointly proposed to increase investments by 10 trillion yuan on infrastructure for cold chain logistics in 2018 (China Consumer Trends, 2018). From the perspective of scholars, research interests focus on optimizing decisions involved in the supply chain for perishable products, including location, inventory and routing problems. Many efforts have been devoted to optimizing those problems simultaneously, which motivates current paper. Still, random supply and demand is the most important issue to tackle regarding the perishable product SCM.

It has been acknowledged that location, inventory and routing problems are three correlated issues in perishable product supply chain management. The problem of locating facilities in perishable product supply chains has been widely addressed. For example, classic location problem such as the Hub Covering Location Problem under Gradual Decay Function are suitable to solve location problem in fresh agricultural products (FAP) supply chain. However, the subject has gotten more attention in recent years due to consumer concerns about food being fresh and healthy. As during the process of transport and inventory operations, the fresh products are exposed to changes in temperature and relative humidity. This is linked to the perishability of the perishable product and its quality at the moment of reaching the final retailer (Orjuela-Castro et al., 2017; Meng, 2016). Efficient inventory management is capable of reducing freight losses for perishable products, thus ensuring the product quality and customer service level (Chaaben et al., 2018; Cooper, 2001). When the perishable products are delivered to lots of customers, it is hard to keep the freshness of delivered food products due to extended travel time and frequent stops to serve customers. Therefore, the efficiency of the vehicle routing is regarded as an important issue to maintain the freshness of customer's perishable products. Efficient vehicle routings are helpful in maximizing the total level of customer satisfaction which depends on the freshness of delivered perishable products (Song and Ko, 2016; Hsu et al., 2007).

Consequently, this research proposes an integrated approach for simultaneously optimizing the decisions involved in the supply chain, typically including location, inventory, and routing strategy. A three-echelon supply chain including suppliers, DCs and retailers is structured in this paper, with the impacts of whether allowing direct shipment from supplier to retailer being considered.

Specifically, the research objectives include:

1) Propose an integrated location-inventory model (LIP) and location-inventory-routing model (LIRP) to quantify the overall costs incurred in the supply chain;
2) Investigate and compare two supply chain structures: allowing direct shipment from supplier to retailer, or not allowing direct shipment from supplier to retailer for perishable products;
3) Develop an efficient solution algorithm to solve the proposed LIP model and LIRP model;
4) Apply the proposed model and algorithm to a real case study.

Before formally moving to the mathematical formulations of the location-inventory and location-inventory-routing problem, here we explain the abbreviations used in this paper: LIP (location-inventory problem), LIRP (location-inventory-routing problem), LIPWDS (LIP considering direct shipment), LIPWNDS (LIP without considering direct shipment), LIRPWDS (LIRP considering direct shipment), LIRPNDS (LIRP without considering direct shipment).

## Literature review

It has been acknowledged that determining an appropriate location for a unit, optimization of inventory strategy, good planning of delivery fleet and vehicle routing, could improve the entire supply chain performance. One of the most important aspects of supply chain management is the integrated decisions on several sub-problems of the entire system. For example, inventory-location problems are the problems in which the integrated solution is achieved by simultaneous optimization of inventory problem and facility location problem. Before formally moving to the integrated LIRP model proposed by this research, the studies dealing with four groups of general decisions on supply chain are reviewed.

### *Location-Inventory problem (LIP)*

A variety of approaches have been adopted by researchers in the field to investigate LIP, including non-linear integer-programming model, set-covering model, etc. (Daskin et al., 2002; Shen et al., 2003; Snyder et al., 2007; Shu et al., 2005). Majority of the researches assumed that there was only one supplier with fixed order lead time and one DC under single-echelon inventory, which can't reflect the real situations. To fill up the gap between the academic research and the reality, researchers attempted to incorporate some practical concerns.

The first practical concern is about the inventory capacity. A mixed integer-programming non-linear model for LIP with deterministic inventory capacity constraint was built (Miranda and Garrido, 2004). Similar work was implemented, considering the stochastic inventory capacity constraint (Miranda and Garrido, 2006). Different from the study by Miranda and Garrido (Miranda and Garrido, 2004; Miranda and Garrido, 2006), the impacts of the variable order lead time, replenishment time and road congestions on the LIP were investigated (Sourirajan et al., 2009). Similar work can be found by Park *et al.*, Romeijn *et al.*, and Vidyarthi *et al.* (Park et al., 2010; Romeijn et al., 2007; Vidyarthi et al., 2007). The second practical concern is about the risk management. The location-inventory problem under risk sharing scheme was studied by Ozsen *et al.* (Ozsen et al., 2008). The impacts of shut-down risks of distribution center were investigated by an inter-programming model based on the supply chain reliability (Chen et al., 2011; Xie et al., 2019). Cui *et al.* proposed a compact mixed integer program formulation to study the reliable un-capacitated fixed charge location problem (Cui et al., 2010). Similar work can be found by Qi *et al.*, Li *et al.*, and Ross *et al.* (Qi and Shen, 2007; Li and Ouyang, 2010; Ross et al., 2008). The third practical concern is about the variety in supply chain network structure. Shahabi *et al.* investigated a three-level LIP based on a mixed integer quadratic program (Shahabi et al., 2014). Escalona *et al.* analyzed the design of a distribution network for fast-moving items under two demand classes (Escalona et al., 2015). Tancrez *et al.* studied a LIP in three-level supply networks (Tancrez et al., 2012).

*Inventory-Routing problem (IRP)*

The inventory routing problem focuses on jointly optimizing inventory replenishment and vehicle routing decisions in a distribution context. Alvarez (2021) investigated an inventory routing problem considering stochastic supply and demands by establishing a two-stage stochastic programming formulation, where routing decisions were made in the first stage, while delivery quantities, inventory levels and specific recourse actions were determined in the second stage (Alvarez et al., 2021). Manousakis (2021) examined the IRP with maximum level inventory policy and proposed a novel two-commodity flow formulation together with a new set of valid inequalities. On this basis, a branch-and-cut algorithm that employed methods for separating various families of cuts was proposed (Manousakis et al., 2021). Schenekemberg et al. (Schenekemberg et al., 2020) targeted at a two-echelon vendor-managed inventory system when a company must make vehicle routing and inventory management decisions, while renting a fleet subject to short- and mid-term agreements. Based on a real-life case in the petrochemical industry, a formulation that took into account vehicle rentals, cleanings, transportation, inventory management, and vehicle returns decisions, was introduced (Schenekemberg et al., 2020). Ji et al. (Ji et al., 2020) proposed a mixed integer linear programming model to minimize the total cost, to solve the inventory routing problem of perishable products with time window constraints. Due to the uncertainty of market demand, the MILP model was further transformed into mixed integer robust programming (MIRP) model by introducing uncertain sets (Ji et al., 2020).

*Location-Routing problem (LRP)*

The location-routing problem is basically a routing problem in which simultaneously optimizing the locations and number of warehouses with vehicle schedules and delivery routes to minimize the entire system cost. Yaghoubi and Akrami (Yaghoubi and Akrami, 2019) investigated the ordering planning of a supply chain with multi suppliers, multi distribution centers, multi customers and one perishable raw material by establishing a mathematical model considering the limitation of raw material corruptibility (perishable material) (Yaghoubi and Akrami, 2019). Navazi et al (Navazi et al., 2019) designed a network for the location-routing problem with two-compartment vehicles responsible for gathering the expired products to the recycling centers in dual-purpose routes by simultaneous pickup and delivery (Navazi et al., 2019). Araghi *et al.* (2021) propoed a mathematical model to find efficient depot location and optimize vehicle routing to transport manufactured goods to depots with the least $CO_2$ pollution possible (Tayebi Araghi et al., 2021). Yu et al. (Yu et al., 2021) investigated the location-routing problem with time-dependent demands (LRPTDD) as an extension of the location-routing problem (LRP) by proposing a mixed-integer nonlinear programming (MINLP) formulation (Yu et al., 2021).

*Location-Inventory-Routing problem (LIRP)*

Shen *et al.* designed an integrated supply chain framework including location, inventory, and routing decisions. However, no specific routing decision was generated from their results (Shen and Qi, 2006). To further obtain the routing schedule, Javid *et al.* for the first time presented a novel model to simultaneously optimize location, allocation and routing (Javid and Azad, 2010). Zhang *et al.* considered a supply chain network with multiple depots (Zhang et al., 2014). Ghorbani *et al.* studied a multi-product and multi-period LIRP in a three-level supply chain (Ghorbani and Jokar, 2016). Tavana *et al.* proposed a multi-echelon logistic network that considering the location of central warehouses (Tavana et al., 2018). Zheng *et al.* studied the integrated optimization problem of LIRP in supply chain network of a motor corporation (Zheng et al., 2019).

However, the above studies considered one-echelon inventory only. It is noticed that multi-echelon inventory is of more practical implications, which has been solved by few scholars. Guerrero investigated a multi-depot, multi-retailer system with storage capacity over a discrete time planning horizon (Guerrero, 2014). Saragih *et al.* studied a LIRP in a three-echelon supply chain system (Saragih et al., 2019).

As for LIRP for perishable products, few literatures can be found. Hiassat *et al.* proposed a model with location, inventory and routing decisions to determine the number and location of required warehouses (Hiassat et al., 2017). Rafie-Majd *et al.* investigated a three-echelon supply chain with a supplier, a number of depots, and a number of retailers (Rafie-Majd et al., 2018). Sebatjane and Adetunji proposed an integrated model for inventory control in a three-echelon supply chain for growing items, with farming, processing and retail echelons. A profit enhancement mechanism which relaxed the traditional zero-ending inventory policy was investigated to test the model effectiveness (Sebatjane and Adetunji, 2021). Esteso *et al.* investigated the impact of product perishability on the design of agri-food supply chains (AFSC). A novel mixed-integer linear programming model was proposed to design entire AFSCs with multiple-products, considering capacity, planting, harvesting, transporting and perishability constraints for a multiple-period horizon (Ana Esteso and Alemany and Ángel Ortiz, 2021). Ma et al. (Ma et al., 2021) developed tripartite game models consisting of the government, one emission-dependent supplier and one retailer in a two-echelon cold chain system, while considering the cap-and-trade regulations and consumers' preference for low-carbon fresh products (Ma et al., 2021). Moubed et al. (Moubed et al., 2021) studied a closed-loop supply chain, and investigated the effects of three strategy groups on the model performance: bargaining, better warehousing and change collection rules (Moubed et al., 2021).

Table 1 summarizes the difference between existing literature and current paper.

This paper aims to fill up the research gap, by integrating the location, inventory and routing decisions under a three-echelon supply chain system for perishable products, considering whether direct shipment from supplier to retailer is allowed or not.

Allowance of delivery from DC to retailer, could minimize the risks and inventory costs incurred in the whole supply chain by sharing the inventory among retailer warehouses. Furthermore, it could utilize the economic scale of DC to reduce the delivery costs. However, this mode might increase the freight losses for perishable products, generated from logistics operations in DC. Especially for the commodity geographically close to the supplier and the commodity on enormous/frequent demand, delivery costs from DC to retailer under this mode might be increased significantly. Consequently, it is necessary to investigate and compare LIP and LIRP for the two scenarios above.

## Methodology

This paper proposes an integrated approach for optimizing location-inventory-routing supply chain for perishable products, with the objective of minimizing the overall costs, including infrastructure construction costs, inventory costs, transportation costs, and freight losses. The decisions to be made are: 1) location decision: to determine and locate a set of candidate DCs and suppliers to open; 2) allocation decision: to identify which retailers are served by each DC or supplier; 3) inventory decision: to determine the optimal order cycle time for DC and retailer; 4) routing decision: to decide the specific delivery routes within the delivery cycle. Fig. 1 explains two delivery modes under a three-echelon supply chain system proposed by this research. Fig. 1(a) indicates that a retailer orders the goods which can be transported from a DC or directly from a supplier. Fig. 1(b) clarifies that a supplier can only deliver goods to the DC.

### *Location-Inventory model*

Before formally moving to the mathematical model, the assumptions and variables are introduced.

#### *Notations, assumptions, and variables*

To simplify the problem, several assumptions are made:

1) One retailer is only served by one DC or one supplier; One DC is only served by one supplier;
2) Supplier doesn't have production capacity;
3) Shortage cost is neglected;
4) Each DC is constrained to dealer inventory capacity;
5) Goods loss rate for any route between two nodes is the same.

The notations and variables used in this paper are listed in Table 2.

#### *Inventory strategy*

This research considers the inventory strategy for both DCs and retailers. Two-echelon inventory system is able to improve the customer service quality by avoiding the stockout risks associated with stochastic demand. This research considers the decentralized and independent inventory strategy, i.e., both DC and retailer which is served by supplier, implement continuous inventory check, and make an order when inventory reaches reorder point. The optimal economic order quantity is determined by EOQ model.

**Table 1**
Summary of existing research.

| Location-inventory problem | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Author (s) | Year | SC Level | Lead Time | Single/Multiple Suppliers | Safety Inventory | Single/Multiple Goods | Inventory Limit | Direct Shipment | Demand Relation |
| Daskin | 2002 | 3 | Constant | Single | Y | Single | N | N | N |
| Shen | 2003 | 3 | Constant | Single | Y | Single | N | N | N |
| Miranda | 2004 | 3 | Constant | Single | Y | Single | Y | N | N |
| Teo | 2004 | 2 | N.A. | Single | N | Single | N | N | N |
| Shu | 2005 | 3 | Constant | Single | Y | Single | N | N | N |
| Vidyarthi | 2007 | 3 | Constant | Multi. | Y | Multi. | Y | N | N |
| Snyder | 2007 | 3 | Constant | Single | Y | Single | N | N | N |
| Romeijn | 2007 | 3 | Constant | Single | Y | Single | N | N | N |
| Sourirajan | 2009 | 3 | Distance related | Single | Y | Single | N | N | N |
| Ozsen | 2010 | 3 | Constant | Single | Y | Single | Y | N | N |
| Park | 2010 | 3 | Distance related | Single | Y | Single | Y | N | N |
| Chen | 2011 | 2 | N.A. | Single | N | Single | N | N | N |
| Tancrez | 2012 | 3 | N.A. | Single | N | Single | Y | Y | N |
| Shahaibi | 2013 | 4 | Constant | Single | Y | Single | N | N | N |
| Yao | 2013 | 3 | Constant | Single | Y | Single | N | N | N |
| Shu | 2014 | 4 | Constant | Single | Y | Single | N | N | N |
| Shahaibi | 2014 | 3 | Constant | Single | Y | Single | Y | N | Y |
| Escalona | 2015 | 3 | Constant | Single | N | Single | N | N | N |
| Shu | 2016 | 3 | Constant | Single | Y | Single | N | N | N |
| Wang | 2016 | 3 | Constant | Single | Y | Single | N | N | N |
| Sun | 2017 | 3 | Constant | Single | Y | Single | N | N | N |
| Guo | 2017 | 3 | Constant | Single | Y | Multi. | Y | N | N |
| Current | 2021 | 3 | Distance related | Single | Y | Multi. | Y | Y | Y |

| Location-inventory-routing problem | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Author (s) | Year | SC Level | Goods Loss | Single/Multiple Period | Single/Multiple Suppliers | Single/Multiple Goods | Time Windows | Direct Shipment |
| Shen | 2007 | 3 | N | Single | Single | Single | N | N |
| Javid | 2010 | 3 | N | Single | Single | Single | N | N |
| Lv | 2011 | 3 | N | Single | Single | Single | Y | N |
| Xing | 2012 | 3 | N | Single | Single | Single | Y | N |
| Cui | 2012 | 3 | N | Multi. | Single | Single | N | N |
| Guo | 2013 | 3 | N | Single | Single | Single | N | N |
| Guerrero | 2014 | 3 | N | Multi. | Single | Single | N | N |
| Nekooghadirli | 2014 | 2 | N | Multi. | Single | Single | N | N |
| Zhang | 2014 | 2 | N | Multi. | Single | Single | N | N |
| Wu | 2016 | 3 | N | Single | Single | Single | N | N |
| Ghorbani | 2016 | 3 | N | Multi. | Mult. | Single | N | N |
| Tavana | 2017 | 4 | Y | Multi. | Single | Single | N | N |
| Rayat | 2017 | 3 | N | Multi. | Single | Single | N | N |
| Yang | 2017 | 3 | N | Single | Single | Single | Y | N |
| Li | 2017 | 3 | N | Single | Single | Single | N | N |
| Tang | 2017 | 3 | N | Single | Single | Single | N | N |
| Wang | 2018 | 3 | N | Single | Single | Single | N | N |
| Rafie-Majd Z | 2018 | 3 | Y | Multi. | Single | Single | Y | N |
| Zhang | 2019 | 4 | N | Single | Single | Single | N | N |
| Indah Saragih | 2019 | 3 | N | Single | Single | Single | N | N |
| Zheng | 2019 | 3 | N | Single | Single | Single | N | N |
| Current paper | 2021 | 3 | Y | Multi. | Single | Single | Y | Y |

(1) DC inventory strategy

Daily demand for retailer's commodities follows normal distribution $(\mu_{fn},\sigma_{fn}^2)$. Considering the interactions among retailers and freight losses, the demand for commodities in DC follows multivariate normal distribution $(D_{fm},U_{fm})$.

$$D_{fm} = \frac{\sum_{n\in I}\mu_{fn}x_{nm}}{1-\theta_f} \tag{1}$$

$$U_{fm} = \sum_{n\in N}\sum_{l\in N}\rho_{nl}\sigma_{fn}\sigma_{fl}x_{nm}x_{lm} \tag{2}$$

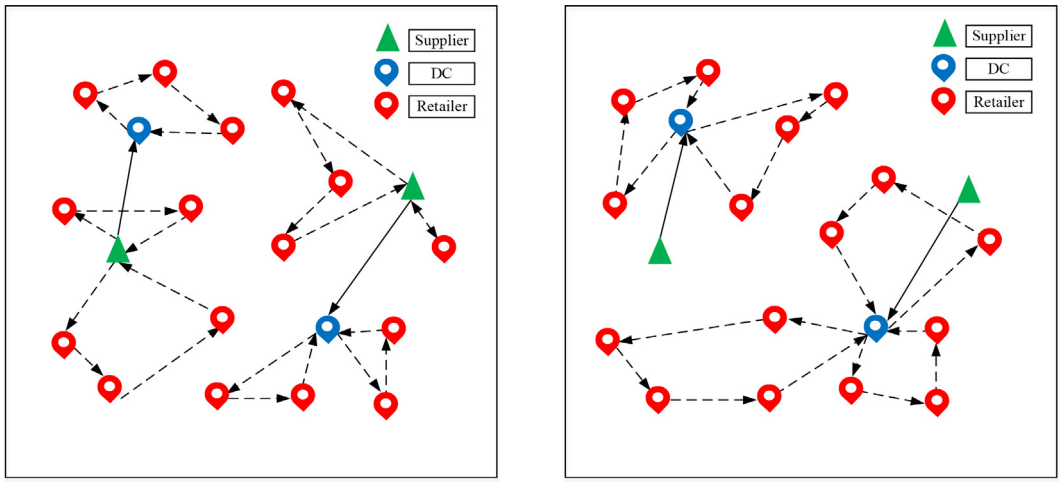



**Fig. 1.** A three-echelon supply chain for LIRPWDS and LIRPNDS.

**Table 2**
Notations and variables used.

| | | |
|---|---|---|
| ***Set*** | | |
| N | Retailers | |
| M | Distribution centers (DCs) | |
| B | Suppliers | |
| F | Goods | |
| ***Parameters*** | | |
| $A_{fm}$ | Ordering cost of commodity *f* of DC *m* per order. | yuan/order |
| $a_{fn}$ | Ordering cost of commodity *f* of retailer *i* per order. | yuan/order |
| $H_{fm}$ | Inventory cost of commodity *f* in DC *m* per unit. | yuan/ (year. pc) |
| $cap_{fm}$ | Dealer inventory capacity of commodity *f* of DC *m*. | yuan/ (year. pc) |
| $h_{fn}$ | Inventory cost of commodity *f* in retailer *n* | yuan/ (year. pc) |
| $l_{fmb}$ | Lead time of commodity *f* of DC *m* from supplier *b*. | day |
| $l_{fnb}$ | Lead time of commodity *f* of retailer *n* from supplier *b*. | day |
| $l_{fnm}$ | Delivery cycle of commodity *f* from DC *m* to retailer *n*. | day |
| $\lambda$ | No. of working days a year. | day |
| $p_b$ | Construction fee of supplier *b*. | day |
| $q_{mb}$ | Annual infrastructure construction fee of DC *m* served by *b*. | yuan |
| $t_{fmb}$ | Unit freight fare of commodity *f* transported from *b* to *m*. | yuan/pc |
| $t_{fnb}$ | Unit freight fare of commodity *f* transported from *b* to *n*. | yuan/pc |
| $t_{fnm}$ | Unit freight fare of commodity *f* transported from *m* to *n*. | yuan/pc |
| $\mu_{fn}$ | Daily demand for commodity *f* of retailer *n*. | pc/day |
| $\sigma_{fn}$ | Standard deviation of$\mu_{fn}$. | |
| $\rho_{nl}$ | Correlation coefficient of demand of *n* and *i*. | |
| $Z_\alpha$ | Percentile Values of $\alpha$ which follows normal distribution. | |
| $\theta_f$ | Freight loss rate of commodity *f*. | |
| $o_f$ | Unit freight loss cost of commodity *f*. | yuan/pc |
| ***Variables*** | | |
| $D_{fm}$ | Demand for commodity *f* of DC *m* per day. | pc/order |
| $U_{fm}$ | Variance of$D_{fm}$. | |
| $L_{fm}$ | Lead time of commodity *f* ordered by DC *m*. | day |
| $L_{fn}$ | Lead time of commodity *f* ordered by retailer *n*. | day |
| $Q_{fm}$ | Optimal order quantity of commodity f of DC m. | pc/day |
| $Q_{fn}$ | Optimal order quantity of commodity f of retailer n. | pc/day |
| $SS_{fm}$ | Safety inventory of commodity *f* of DC *m*. | pc |
| $SS_{fn}$ | Safety inventory of commodity *f* of retailer *n*. | pc |
| $R_{fm}$ | Reorder point of commodity *f* of DC *m*. | pc |
| $R_{fn}$ | Reorder point of commodity *f* of retailer *n*. | pc |
| ***Decision Variables*** | | |
| $x_{nm}$ | 1 if retailer *n* is served by DC *m*; 0 otherwise. | |
| $y_{mb}$ | 1 if DC *m* is served by supplier *b*; 0 otherwise. | |
| $z_{nb}$ | 1 if retailer *n* is served by supplier *b*; 0 otherwise. | |
| $w_b$ | 1 if supplier *b* is open; 0 otherwise. | |

Lead time of a DC is:

$$L_{fm} = \sum_{b\in B} l_{fmk} y_{mk} \tag{3}$$

Safety inventory of a DC is:

$$SS_{fm} = Z_\alpha \sqrt{U_{fm} L_{fm}} = Z_\alpha \sqrt{\sum_{n\in N}\sum_{l\in N}\sum_{b\in B} \rho_{nl}\sigma_{fn}\sigma_{fl} l_{fmk} x_{nm} x_{lm} y_{mb}} \tag{4}$$

Reorder point of a distribution center is:

$$\begin{aligned} R_{fj} &= SS_{fm} + L_{fm} D_{fm} \\ &= Z_\alpha \sqrt{\sum_{n\in N}\sum_{l\in N}\sum_{b\in B} \rho_{nl}\sigma_{fn}\sigma_{fl} l_{fmb} x_{nm} x_{lm} y_{mb}} + \frac{\sum_{n\in N}\sum_{b\in B} \mu_{fn} l_{fmb} x_{nm} y_{mb}}{1-\theta_f} \end{aligned} \tag{5}$$

The overall inventory of a DC includes the dealer inventory and safety inventory:

$$INV_{f\mathrm{m}} = \frac{Q_{fm}}{2} + SS_{fm} = \frac{Q_{fm}}{2} + Z_\alpha \sqrt{\sum_{n\in N}\sum_{l\in N}\sum_{b\in B} \rho_{nl}\sigma_{fn}\sigma_{fl} l_{fmb} x_{nm} x_{lm} y_{mb}} \tag{6}$$

(2) Retailer inventory strategy

In the supply chain, a retailer is the final point connecting with customers. Hence, its inventory strategy might influence the service level of the retailers and even the whole supply chain. In this research, a retailer orders commodities either from a DC or directly from a supplier. When the retailer is served by a DC, its safety stock is located in the DC. The retailer has dealer inventory only. With delivery cycle$\sum_{m\in M} l_{fnm} x_{nm}$, the dealer inventory is$\mu_{fn}\sum_{m\in M} l_{fnm} x_{nm}/2$. When the retailer is served by a supplier, the safety inventory and dealer inventory need to be considered for the retailer. With the lead time$L_{fn} = \sum_{b\in B} l_{fnb} z_{nb}$, the retailer's safety inventory is:

$$SS_{fn} = Z_\alpha \sigma_{fn} \sqrt{L_{fn}} = Z_\alpha \sigma_{fn} \sqrt{\sum_{b\in B} l_{fnb} z_{nb}} \tag{7}$$

The retailer's overall inventory is:

$$INV_{fn} = \frac{\mu_{fn}\sum_{m\in M} l_{fnm} x_{nm}}{2} + \frac{Q_{fn}}{2} + Z_\alpha \sigma_{fn} \sqrt{\sum_{b\in B} l_{fnb} z_{nb}} \tag{8}$$

Reorder point of a retailer is:

$$R_{fn} = SS_{fn} + L_{fn}\mu_{fn} = Z_\alpha \sigma_{fn} \sqrt{\sum_{b\in B} l_{fnb} z_{nb}} + \sum_{b\in B} l_{fnb}\mu_{fn} z_{nb} \tag{9}$$

*Supply chain cost analysis*

The costs involved in the whole supply chain include the infrastructure construction costs, transportation costs, inventory costs and freight loss costs.

(1) Infrastructure Construction Cost.

$$FIX = \sum_{b\in B} p_b w_b + \sum_{m\in M}\sum_{b\in B} q_{mb} y_{mb} \tag{10}$$

(2) Transportation Cost.

The transportation costs include the main-line transportation (from a supplier to a DC) and last-mile delivery costs (from a supplier/DC to a retailer). Given the overall demand in a DC$\sum_{f\in F}\sum_{m\in M}\sum_{b\in M}\lambda D_{fm} y_{fm}$, the annual demand for all commodities of a DC is$\sum_{f\in F}\sum_{m\in M}\sum_{b\in M}\lambda D_{fm} y_{fm}$. Considering the freight losses incurred in transportation between any two points, the annual freight volume ordered from a supplier to a retailer is$\sum_{f\in F}\sum_{m\in M}\sum_{b\in B}\lambda D_{fm} y_{fm}/(1-\theta_f)$. Similarly, the annual freight volume from a DC to a retailer is $\sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\lambda\mu_{fn} x_{nm}/(1-\theta_f)$ and the annual freight volume from a supplier to a retailer is$\sum_{f\in F}\sum_{n\in N}\sum_{b\in B}\lambda\mu_{fn} z_{nb}/(1-\theta_f)$. Hence, the overall transportation costs incurred in the whole supply chain are:

$$TRA = \sum_{f\in F}\sum_{i\in I}\sum_{j\in J}\sum_{k\in K} \frac{\lambda\mu_{fn} t_{fmb} x_{nm} y_{mb}}{(1-\theta_f)^2} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M} \frac{\lambda\mu_{fn} t_{fnm} x_{nm}}{1-\theta_f} + \sum_{f\in F}\sum_{n\in N}\sum_{b\in B} \frac{\lambda\mu_{fn} t_{fnb} z_{nb}}{1-\theta_f} \tag{11}$$

(3) Inventory Cost

The overall inventory costs include the ordering cost and the inventory carrying cost, which is composed of dealer inventory cost and safety inventory cost.

$$H_{fm}INV_{fm} = \frac{Q_{fm}H_{fm}}{2} + Z_{\alpha}H_{fm}\sqrt{\sum_{n\in N}\sum_{l\in N}\sum_{b\in B}\rho_{nl}\sigma_{fn}\sigma_{fl}l_{fmb}x_{nm}x_{lm}y_{mb}} \tag{12}$$

Order cost of a DC is:

$$A_{fm}\frac{\lambda D_{fm}}{Q_{fm}} = \frac{\lambda A_{fm}\sum_{n}\mu_{fn}x_{nm}}{Q_{fm}(1-\theta_f)} \tag{13}$$

A retailer's inventory costs include the dealer inventory cost of a retailer served by a DC, and the dealer inventory, safety stock and order costs of a retailer served by a supplier:

$$h_{fn}\frac{\mu_{fn}\sum_{m\in M}l_{fnm}x_{nm}}{2} + \frac{Q_{fn}h_{fn}}{2} + Z_{\alpha}\sigma_{fn}h_{fn}\sqrt{\sum_{b\in B}l_{fnb}z_{nb}} + \frac{\lambda A_{fn}\mu_{fn}\sum_{b\in B}z_{nb}}{Q_{fn}(1-\theta_f)} \tag{14}$$

Consequently, the overall inventory costs are:

$$\begin{aligned} INV = {} & \sum_{f\in F}\sum_{m\in M}\frac{Q_{fm}H_{fm}}{2} + \sum_{f\in F}\sum_{m\in M}Z_{\alpha}H_{fm}\sqrt{\sum_{n\in N}\sum_{l\in N}\sum_{b\in M}\rho_{nl}\sigma_{fn}\sigma_{fl}l_{fmb}x_{nm}x_{lm}y_{mb}} + \sum_{f\in F}\sum_{m\in M}\frac{\lambda A_{fm}\sum_{n}\mu_{fn}x_{nm}}{Q_{fm}(1-\theta_f)} \\ & + \sum_{f\in F}\sum_{n\in N}\frac{Q_{fn}h_{fn}}{2} + \sum_{f\in F}\sum_{n\in N}\frac{h_{fn}\mu_{fn}\sum_{m\in M}l_{fnm}x_{nm}}{2} + \sum_{f\in F}\sum_{n\in N}Z_{\alpha}\sigma_{fn}h_{fn}\sqrt{\sum_{b\in B}l_{fnb}z_{nb}} + \sum_{f\in F}\sum_{n\in N}\frac{\lambda A_{fn}\mu_{fn}\sum_{b\in B}z_{nb}}{Q_{fn}(1-\theta_f)} \end{aligned} \tag{15}$$

(4) Freight Losses

For a DC, the annual demand for all commodities is$\sum_{f\in F}\sum_{j\in J}\sum_{k\in K}\lambda D_{fj}y_{fj}$. Considering the freight loss incurred from transportation process, the actual demand for a DC from a supplier is$\sum_{f\in F}\sum_{j\in J}\sum_{k\in K}\lambda D_{fj}y_{fj}/(1-\theta_f)$. Similarly, for a retailer, the annual demand for all commodities is$\sum_{f\in F}\sum_{i\in I}\lambda\mu_{fi}$. The actual demand for a retailer from a DC is $\sum_{f\in F}\sum_{i\in I}\sum_{j\in J}\lambda\mu_{fi}x_{ij}/(1-\theta_f)$ and the actual demand for a retailer from a supplier is$\sum_{f\in F}\sum_{i\in I}\sum_{k\in K}\lambda\mu_{fi}z_{ik}/(1-\theta_f)$. Consequently, the freight losses are:

$$\begin{aligned} DET &= \sum_{f\in F}\sum_{m\in M}\sum_{b\in M}\frac{\lambda o_f\theta_f D_{fm}y_{mb}}{1-\theta_f} + \sum_{f\in F}\sum_{m\in M}\lambda o_f\theta_f D_{fm} + \sum_{f\in F}\sum_{n\in N}\sum_{b\in B}\frac{\lambda o_f\theta_f\mu_{fn}z_{nb}}{1-\theta_f} \\ &= \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\sum_{b\in M}\frac{\lambda o_f\theta_f\mu_{fn}x_{nm}y_{mb}}{(1-\theta_f)^2} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\frac{\lambda o_f\theta_f\mu_{fn}x_{nm}}{1-\theta_f} + \sum_{f\in F}\sum_{n\in N}\sum_{b\in B}\frac{\lambda o_f\theta_f\mu_{fn}z_{nb}}{1-\theta_f} \end{aligned}$$

*Model formulation for LIP considering direct shipment (LIPWDS)*

The proposed LIP model is to minimize the overall costs incurred in the proposed two-echelon supply chain.

$$\begin{aligned} Min\ & \sum_{b\in B}p_b w_b + \sum_{m\in M}\sum_{b\in B}q_{mb}y_{mb} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\sum_{b\in B}\frac{\lambda\mu_{fn}t_{fmb}x_{nm}y_{mb}}{(1-\theta_f)^2} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\frac{\lambda\mu_{fn}t_{fnm}x_{nm}}{1-\theta_f} \\ & + \sum_{f\in F}\sum_{n\in N}\sum_{b\in B}\frac{\lambda\mu_{fn}t_{fnb}znb}{1-\theta_f} + \sum_{f\in F}\sum_{n\in N}Z_{\alpha}H_{fj}\sqrt{\sum_{n\in N}\sum_{l\in N}\sum_{b\in B}\rho_{nl}\sigma_{fn}\sigma_{fl}l_{fmb}x_{nm}x_{lm}y_{mb}} \\ & + \sum_{f\in F}\sum_{m\in M}\frac{\lambda A_{fm}\sum_{n}\mu_{fn}x_{nm}}{Q_{fm}(1-\theta_f)} + \sum_{f\in F}\sum_{m\in M}\frac{Q_{fm}H_{fm}}{2} + \sum_{f\in F}\sum_{n\in N}\frac{Q_{fn}h_{fn}}{2} + \sum_{f\in F}\sum_{n\in N}\frac{h_{fn}\mu_{fn}\sum_{m\in M}l_{fnm}x_{nm}}{2} \\ & + \sum_{f\in F}\sum_{n\in N}Z_{\alpha}\sigma_{fn}h_{fn}\sqrt{\sum_{b\in B}l_{fnb}z_{nb}} + \sum_{f\in F}\sum_{n\in N}\frac{\lambda A_{fn}\mu_{fn}\sum_{b\in B}z_{nb}}{Q_{fn}(1-\theta_f)} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\sum_{b\in B}\frac{\lambda o_f\theta_f\mu_{fn}x_{nm}y_{mb}}{(1-\theta_f)^2} \\ & + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\frac{\lambda o_f\theta_f\mu_{fn}x_{nm}}{1-\theta_f} + \sum_{f\in F}\sum_{n\in N}\sum_{b\in B}\frac{\lambda o_f\theta_f\mu_{fn}z_{nb}}{1-\theta_f} \end{aligned} \tag{17}$$

To differentiate $Q_{fn}$ and $Q_{fm}$ respectively, the optimal order quantity for a retailer and a DC is:

$$Q_{fn}^{*} = \left(\frac{2\lambda a_{fn}\mu_{fn}\sum_{b\in B}z_{nb}}{h_{fn}(1-\theta_f)}\right)^{1/2}, Q_{fm}^{*} = \left(\frac{2\lambda A_{fn}\sum_{n\in N}\mu_{fn}x_{nm}}{H_{fn}(1-\theta_f)}\right)^{1/2} \tag{18, 19}$$

Replace $Q_{fn}^{*}$ and $Q_{fm}^{*}$ in the objective function, then:

$$
\begin{aligned}
Min\ & \sum_{b\in B} p_b w_b + \sum_{m\in M}\sum_{b\in B} q_{mb} y_{mb} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\sum_{b\in B} \frac{\lambda \mu_{fn} t_{fmb} x_{nm} y_{mb}}{\left(1-\theta_f\right)^2} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M} \frac{\lambda \mu_{fn} t_{fnm} x_{nm}}{1-\theta_f} \\
& + \sum_{f\in F}\sum_{n\in N}\sum_{b\in B} \frac{\lambda \mu_{fn} t_{fnb} z_{nb}}{1-\theta_f} + \sum_{f\in F}\sum_{m\in M} \sqrt{\frac{2\lambda A_{fm} H_{fm} \sum_{n\in N} \mu_{fn} x_{nm}}{1-\theta_f}} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M} \frac{h_{fn} \mu_{fn} l_{fnm} x_{nm}}{2} \\
& + \sum_{f\in F}\sum_{m\in M} Z_\alpha H_{fm} \sqrt{\sum_{n\in N}\sum_{l\in N}\sum_{b\in B} \rho_{nl} \sigma_{fn} \sigma_{fl} l_{fmb} x_{nm} x_{lm} y_{mb}} + \sum_{f\in F}\sum_{n\in N} \sqrt{\frac{2\lambda a_{fn} h_{fn} \mu_{fn} \sum_{b\in B} z_{nb}}{1-\theta_f}} \\
& + \sum_{f\in F}\sum_{n\in N} Z_\alpha \sigma_{fn} h_{fn} \sqrt{\sum_{b\in B} l_{fnb} z_{nb}} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\sum_{b\in M} \frac{\lambda o_f \theta_f \mu_{fn} x_{nm} y_{mb}}{\left(1-\theta_f\right)^2} \\
& + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M} \frac{\lambda o_f \theta_f \mu_{fn} x_{nm}}{1-\theta_f} + \sum_{f\in F}\sum_{n\in N}\sum_{b\in B} \frac{\lambda o_f \theta_f \mu_{fn} z_{nb}}{1-\theta_f}
\end{aligned} \tag{20}
$$

S.T.

$$y_{mb} \leqslant w_b, \quad \forall m \in M, b \in B \tag{21}$$

$$z_{nb} \leqslant w_b, \quad \forall n \in N, b \in B \tag{22}$$

$$x_{nm} \leqslant \sum_{b\in B} y_{mb}, \quad \forall n \in N, m \in M \tag{23}$$

$$\sum_{n\in N} \mu_{fn} x_{nm} \leqslant Cap_{fm}, \quad \forall f \in F, m \in M \tag{24}$$

$$\sum_{n\in N} x_{nm} + \sum_{b\in B} z_{nb} = 1, \quad \forall n \in N \tag{25}$$

$$\sum_{b\in B} y_{mb} \leqslant 1, \quad \forall m \in M \tag{26}$$

$$x_{nm}, y_{mb}, z_{nb}, w_b \in \{0,1\} \quad \forall n \in N, m \in M, b \in B \tag{27}$$

Eq. (20) is the objective function. Eq. (21, 22) ensure that only the selected supplier is able to serve. Eq. (23) indicate that only the selected DC can serve. Eq. (24) limits the deal inventory of a DC. Eq. (25) promises that a retailer can be only served by one DC or one supplier. Eq. (26) suggests that a DC is served by one supplier at most. Eq. (27) defines the 0–1 variable.

*Model formulation for LIP without considering direct shipment (LIPWNDS)*

$$
\begin{aligned}
Min\ & \sum_{b\in B} p_b w_b + \sum_{m\in M}\sum_{b\in B} q_{mb} y_{mb} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\sum_{b\in M} \frac{\lambda \mu_{fn} t_{fmb} x_{nm} y_{mb}}{\left(1-\theta_f\right)^2} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M} \frac{\lambda \mu_{fn} t_{fnm} x_{nm}}{1-\theta_f} \\
& + \sum_{f\in F}\sum_{m\in M} \sqrt{\frac{2\lambda A_{fm} H_{fm} \sum_{n\in N} \mu_{fn} x_{nm}}{1-\theta_f}} + \sum_{f\in F}\sum_{m\in M} Z_\alpha H_{fm} \sqrt{\sum_{n\in N}\sum_{l\in N}\sum_{n\in B} \rho_{nl} \sigma_{fn} \sigma_{fl} l_{fmb} x_{nm} x_{lm} y_{mb}} \\
& + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M} \frac{h_{fn} \mu_{fn} l_{fnm} x_{nm}}{2} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\sum_{b\in B} \frac{\lambda o_f \theta_f \mu_{fn} x_{nm} y_{mb}}{\left(1-\theta_f\right)^2} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M} \frac{\lambda o_f \theta_f \mu_{fn} x_{nm}}{1-\theta_f}
\end{aligned} \tag{28}
$$

S.T.

$$y_{mb} \leqslant w_b, \quad \forall m \in M, b \in B \tag{29}$$

$$x_{nm} \leqslant \sum_{b\in B} y_{mb}, \quad \forall n \in N, m \in M \tag{30}$$

$$\sum_{n\in N} \mu_{fn} x_{nm} \leqslant Cap_{fm}, \quad \forall f \in F, m \in M \tag{31}$$

$$\sum_{n\in N} x_{nm} = 1, \quad \forall n \in N \tag{32}$$

$$\sum_{b\in B} y_{mb} \leqslant 1, \quad \forall m \in M \tag{33}$$

$$x_{nm}, y_{mb}, w_b \in \{0,1\} \quad \forall n \in N, m \in M, b \in B \tag{34}$$

Eq. (29) ensures that only the selected supplier is able to serve. Eq. (30) indicate that only the selected DC can serve. Eq. (31) limits the deal inventory of a DC. Eq. (32) promises that a retailer can be only served by one DC or one supplier. Eq. (33) suggests that a DC is served by one supplier at most. Eq. (34) defines the 0–1 variable.

*Solution algorithm*

The proposed two LIP models are mixed integer nonlinear programming models, which are NP-hard. To solve the models, this research firstly linearizes the objective function of both LIPWDS and LIPWNDS, secondly transforms the models into mixed integer quadratic programming models, and thirdly solves the models using a commercial software package CPLEX programmed by GAMS. The solving process of LIPWDS is explained in details here.

(1) Linearization of objective function

Here introduce a binary variable $M_{nmb}$ to replace the quadratic term $x_{nm}y_{mb}$ in the objective function. Two constraints are introduced.

$$\sum_{b\in B} M_{nmb} = x_{nm} \quad \forall n \in N, m \in M \tag{35}$$

$$M_{nmb} \leqslant y_{mb} \quad \forall n \in N, m \in M, b \in B \tag{36}$$

$x_{nm}y_{mb} = 1$ if and only if$x_{nm} = y_{mb} = 1$. $M_{nmb} = 1$ if and only if$x_{nm} = y_{mb} = 1$. Hence we replace $x_{nm}y_{mb}$ with$M_{nmb}$. Since$x_{nm}$, $y_{mb}$ and $M_{nmb}$ are binary variables, we replace them with their squares, i.e.,$x_{nm}^2 = x_{nm}[0, 1]$, $y_{mb}^2 = y_{mb}$ and$M_{nmb}^2 = M_{nmb}$.

(2) Transformation to mixed integer quadratic programming models

Here we introduce four auxiliary variables$v_{1fm}, v_{2fm}\, v_{3fn}$ and$v_{4fn}$. The model is converted to a mixed integer quadratic programming model:

$$\begin{aligned} Min\ & \sum_{b\in B} p_b w_b + \sum_{m\in M}\sum_{b\in B} q_{mb} y_{mb} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\sum_{b\in B} \frac{\lambda \mu_{fn} t_{fmb} M_{nmb}}{\left(1-\theta_f\right)^2} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M} \frac{\lambda \mu_{fn} t_{fnm} x_{nm}}{1-\theta_f} \\ & + \sum_{f\in F}\sum_{n\in N}\sum_{b\in B} \frac{\lambda \mu_{fn} t_{fnb} z_{nb}}{1-\theta_f} + \sum_{f\in F}\sum_{m\in M} v_{1fm} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M} \frac{h_{fn}\mu_{fn} l_{fnm} x_{nm}}{2} + \sum_{f\in F}\sum_{m\in M} Z_\alpha H_{fm}\, v_{2fm} + \sum_{f\in F}\sum_{n\in N} v_{3fn} \\ & + \sum_{f\in F}\sum_{n\in N} Z_\alpha \sigma_{fn} h_{fn}\, v_{4fn} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M}\sum_{b\in B} \frac{\lambda o_f \theta_f \mu_{fn} M_{nmb}}{\left(1-\theta_f\right)^2} + \sum_{f\in F}\sum_{n\in N}\sum_{m\in M} \frac{\lambda o_f \theta_f \mu_{fn} x_{nm}}{1-\theta_f} + \sum_{f\in F}\sum_{n\in N}\sum_{b\in B} \frac{\lambda o_f \theta_f \mu_{fn} z_{nb}}{1-\theta_f} \end{aligned}$$

S.T.

$$y_{mb} \leqslant w_b, \quad \forall m \in M, b \in B \tag{38}$$

$$z_{nb} \leqslant w_b, \quad \forall n \in N, b \in B \tag{39}$$

$$x_{nm} \leqslant \sum_{b\in B} y_{mb}, \quad \forall n \in N, m \in M \tag{40}$$

$$\sum_{n\in N} \mu_{fn} x_{nm} \leqslant Cap_{fm}, \quad \forall f \in F, m \in M \tag{41}$$

$$\sum_{n\in N} x_{nm} + \sum_{b\in B} z_{nb} = 1, \quad \forall n \in N \tag{42}$$

$$\sum_{b\in B} y_{mb} \leqslant 1, \quad \forall m \in M \tag{43}$$

$$\frac{2\lambda A_{fm} H_{fm} \sum_{n\in N} \mu_{fn} x_{nm}}{1-\theta_f} \leqslant v_{1fm}^2 \quad \forall f \in F, m \in M \tag{44}$$

$$\sum_{n\in N}\sum_{l\in N}\sum_{b\in B} \rho_{nl}\sigma_{fn}\sigma_{fl} l_{fmb} M_{nmb} M_{lmb} \leqslant v_{2fm}^2 \quad \forall f \in F, m \in M \tag{45}$$

$$\frac{2\lambda a_{fn} h_{fn} \mu_{fn} \sum_{b\in B} z_{nb}}{1-\theta_f} \leqslant v_{3fn}^2 \quad \forall f \in F, n \in N \tag{46}$$

$$\sum_{b\in B} l_{fnb} z_{nb} \leqslant v_{4fn}^2 \quad \forall f \in F, n \in N \tag{47}$$

$$v_{1fm},\ v_{2fm},\ v_{3fn},\ v_{4fn} \geqslant 0, \quad \forall f \in F, n \in N, m \in M \tag{48}$$

$$x_{nm}, y_{mb}, z_{nb}, w_b, M_{nmb} \in \{0,1\} \quad \forall n \in N, m \in M, b \in B \tag{49}$$

*Numerical example*

A variety of numerical examples are proposed here to investigate and compare the costs incurred in LIPWDS and LIPWNDS. Table 3 presents the proposed test networks with $s$ as number of suppliers, $n$ as number of DCs, $m$ as number of retailers and $b$ as types of fresh products. Setting of parameters are inspired by Park (2010). $\bar{\mu}_f$ represents the daily demand for commodity $f$; $\tau$ is the floating percentage; $\bar{c}_1$ and $\bar{c}_0$ are the unit transportation cost from supplier to DC, and from supplier to retailer, respectively. The value for each parameter is: $V_1 = 0.5$, $V_2 = V_3 = V_1 + 0.5$, $\mu_0 = 10$, $\sigma_0^2$=6, $C_0 = 8$, $h_0 = 3$, $A_0 = 10$, $\tau = 0.5$, $q_0 = 5$, $\lambda = 120$.

The geographical distribution of suppliers, DCs and retailers is randomly generated within a designated area.

(1) Cost analysis for LIPWDS model

Fig. 2 presents the sharing of freight loss cost, inventory cost, transportation cost and construction cost over the total costs. It is found that the transportation cost is the most dominant component, accounting for 41%-58% of the overall supply chain cost for 11 tested networks. Infrastructure construction cost and inventory cost come in at 16%-26%, 10%-24% of the overall costs, respectively. Freight loss cost shares 7%-16% of the overall. It is then concluded that reduction in transportation cost will be very helpful in reducing the overall supply chain cost.

Fig. 3 suggests that under LIPWDS model, retailer's inventory cost accounts for 89% of the overall inventory cost on average, whereas the inventory cost accounts for 11%. Obviously, under LIPWDS model, retailers are under high pressure of inventory cost control. It is then of great importance to upgrade the warehousing infrastructure and improve the efficiency of ordering system in order to reduce the inventory cost.

The details of inventory strategy adopted by retailer and DC is explained in Table 4. The retailers served by DC tend to utilize centralized inventory strategy, i.e., DC orders the goods and keeps safety stock. The optimal order quantity of a retailer

**Table 3**
Test network and parameters.

| Test Network | PB1 | PB2 | PB3 | PB4 | PB5 | PB6 | PB7 | PB8 | PB9 | PB10 | PB11 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $b$ | 3 | 5 | 7 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |
| $s$ | 3 | 3 | 3 | 5 | 5 | 5 | 7 | 7 | 9 | 9 | 9 |
| $n$ | 5 | 5 | 5 | 5 | 7 | 10 | 10 | 13 | 13 | 15 | 15 |
| $m$ | 15 | 20 | 25 | 25 | 25 | 30 | 35 | 40 | 40 | 40 | 50 |
| Parameters related to inventory | | | | | | | | | | Other parameters | |
| $H_{fi} = h_0 \cdot (1 + U[-\tau, \tau])$ | | | | | | | | | | $\mu_{fi} = \mu_0 \cdot U[1,5]$ | |
| $h_{fi} = (h_0 + 1) \cdot (1 + U[-\tau, \tau])$ | | | | | | | | | | $\sigma_{fi} = (\sigma_0^2 \cdot U[1,5])^{1/2}$ | |
| $l_{fjk} = 3 \cdot U[1,5]$ | | | | | | | | | | $\theta_f = U[0.05, 0.1]$ | |
| $l_{fik} = 3 \cdot U[1,5]$ | | | | | | | | | | $O_f = U[5,10]$ | |
| $l_{fij} = U[1,5]$ | | | | | | | | | | $Cap_{fj} = 3 \cdot C_0 \cdot \bar{\mu}_f \cdot (1 + U[-\tau, \tau])$ | |
| $A_{fj} = A_0 \cdot \bar{c}_0 \cdot \bar{\mu}_f \cdot (1 + U[-\tau, \tau])$ | | | | | | | | | | $q_{jk} = q_0 \cdot b \cdot \bar{c}_1 \cdot \sum Cap_{fj} \cdot (1 + U[-\tau, \tau])$ | |
| $A_{fi} = (A_0 + 2) \cdot c_1 \cdot \bar{\mu}_f \cdot (1 + U[-\tau, \tau])$ | | | | | | | | | | $p_k = 3 \cdot \bar{q} \cdot (1 + U[-\tau, \tau])$ | |

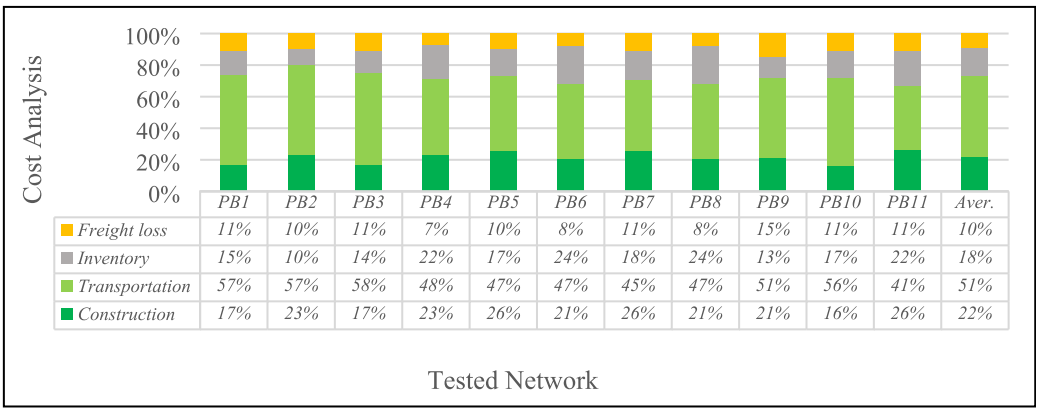



**Fig. 2.** Proportion of cost considering direct shipment.

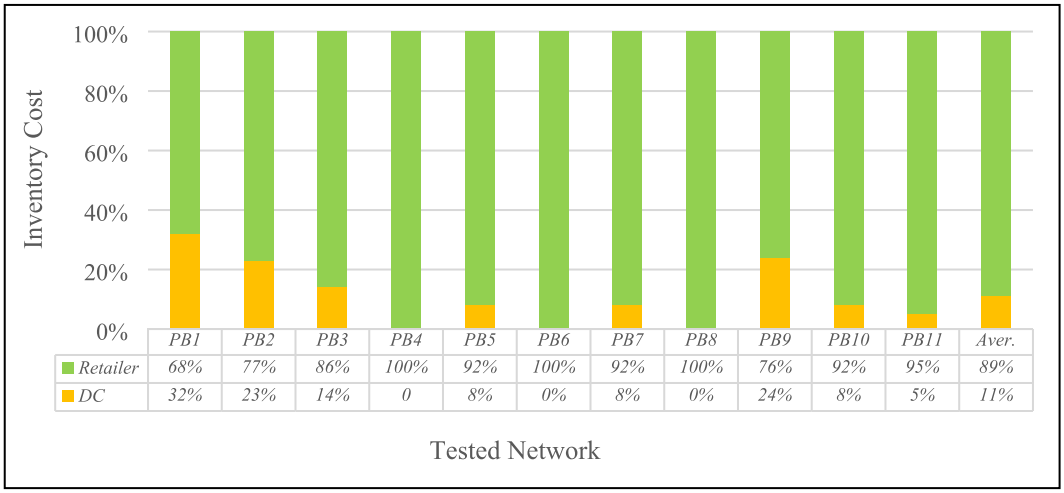



**Fig. 3.** Proportion of inventory cost considering direct shipment.

is less than that of the retailer served by the supplier. For example, for network PB1, the optimal order quantity for commodity A of DC D1 is 2449. D1 serves 6 retailers. The average optimal order quantity is 408 for each retailer. At the same time, when the retailers are served by suppliers, the optimal order quantity for commodity A is between 1315 and 2640. It indicates the superiority of centralized inventory strategy over the decentralized strategy.

(2) Cost analysis for LIPWNDS model

Fig. 4 suggests that transportation cost accounts for 44%-58% of the overall supply chain cost. The second largest component is infrastructure construction cost, come in at 23%-26% of the overall. Inventory cost accounts for 4%-5% of overall. For the LIPWNDS model, all the orders are transported from DCs which hold the safety stock. This centralized inventory strategy is effective in reducing the inventory cost. It is then important to reduce the transportation costs incurred in the LIPWNDS network.

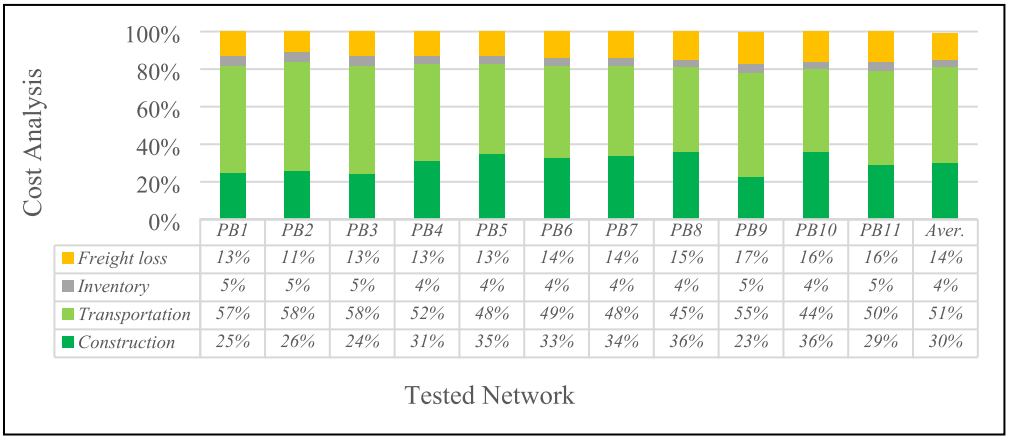
To further investigate the inventory costs of LIPWNDS model, Fig. 5 demonstrates that DC's inventory cost accounts for 88% of overall inventory costs whereas the retailer's inventory shares 12% of overall. It is then of great importance to reduce the inventory cost of DC in order to reduce the whole supply chain costs.

The computational results indicate that for all the proposed networks, the overall costs incurred in the LIPWDS model are less than those of LIPWNDS model. The average reductions achieved by LIPWDS are 10.6%.

**Table 4**
Inventory strategy of DC and retailer considering direct shipment for PB1.

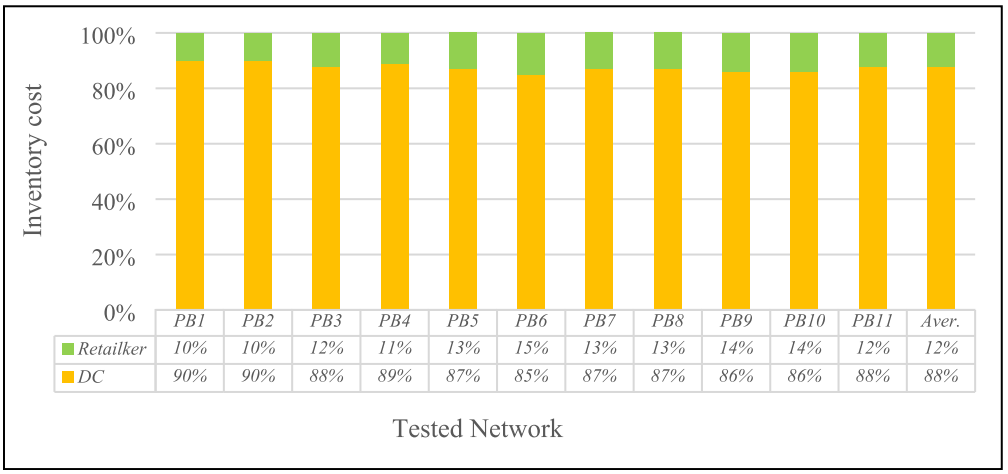
| Inventory strategy of DC served by supplier directly for network PB1 | | | |
|---|---|---|---|
| Commodity | Inventory Strategy | D1 | D4 |
| A | Re-order point | 2359 | 987 |
| | Optimal order quantity | 2449 | 2074 |
| | Safety stock | 136 | 56 |
| B | Re-order point | 1522 | 1338 |
| | Optimal order quantity | 2966 | 1618 |
| | Safety stock | 104 | 56 |
| C | Re-order point | 1125 | 1571 |
| | Optimal order quantity | 1727 | 1981 |
| | Safety stock | 103 | 82 |

| Inventory strategy of retailer served by supplier directly for network PB1 | | | | | | | |
|---|---|---|---|---|---|---|---|
| | Inventory Strategy | R2 | R3 | R7 | R8 | R10 | R12 |
| A | Re-order point | 341 | 632 | 391 | 187 | 228 | 396 |
| | Optimal order quantity | 1988 | 2640 | 1315 | 1716 | 1871 | 1757 |
| | Safety stock | 20 | 31 | 26 | 13 | 17 | 22 |
| B | Re-order point | 440 | 366 | 314 | 126 | 175 | 135 |
| | Optimal order quantity | 2613 | 2081 | 1505 | 2104 | 1059 | 1223 |
| | Safety stock | 22 | 15 | 19 | 13 | 24 | 17 |
| C | Re-order point | 204 | 558 | 158 | 382 | 85 | 160 |
| | Optimal order quantity | 1290 | 2589 | 1761 | 1317 | 1264 | 1650 |
| | Safety stock | 21 | 20 | 12 | 18 | 15 | 16 |




| | PB1 | PB2 | PB3 | PB4 | PB5 | PB6 | PB7 | PB8 | PB9 | PB10 | PB11 | Aver. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Freight loss | 13% | 11% | 13% | 13% | 13% | 14% | 14% | 15% | 17% | 16% | 16% | 14% |
| Inventory | 5% | 5% | 5% | 4% | 4% | 4% | 4% | 4% | 5% | 4% | 5% | 4% |
| Transportation | 57% | 58% | 58% | 52% | 48% | 49% | 48% | 45% | 55% | 44% | 50% | 51% |
| Construction | 25% | 26% | 24% | 31% | 35% | 33% | 34% | 36% | 23% | 36% | 29% | 30% |

**Fig. 4.** Proportion of cost without considering direct shipment.




| | PB1 | PB2 | PB3 | PB4 | PB5 | PB6 | PB7 | PB8 | PB9 | PB10 | PB11 | Aver. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Retailker | 10% | 10% | 12% | 11% | 13% | 15% | 13% | 13% | 14% | 14% | 12% | 12% |
| DC | 90% | 90% | 88% | 89% | 87% | 85% | 87% | 87% | 86% | 86% | 88% | 88% |

**Fig. 5.** Proportion of inventory cost without considering direct shipment.

*Sensitivity analysis*

Based on PB1 network, this paper undertakes sensitivity analysis on unit ordering cost, unit inventory cost, and freight loss rate.

(1) Sensitivity Analysis of Unit Ordering Cost

This research adoptes a variety of $A_0$ (ratio of assumed unit ordering cost to the benchmark 10 yuan/time) to invstigate the impacts of unit ordering costs on the final results.

Table 5 suggests that when $A_0$ increases from 60% to 240%, the reductions of overall costs achieved by LIPWDS model compared with those of LIPWNDS model decrease from 8.03% to 2.44%. Consequently, when the unit ordering cost is relatively small, the LIPWDS model is more benefical in reducing the supply chain cost. However, with the increased unit ordering cost, the increment of invetory costs incurred in the LIPWDS model is greater than that of LIPWNDS. It is then concluded that with the dramatic developments of IT which leads to the better coordination of all supply chain players, it is possible to reduce the unit ordering cost and there is great developemt potential for LIPWDS in the future.

(2) Sensitivity Analysis of Unit Inventory Cost

This research implements a sensitivity analysis by adopting a variety of $h_0$ (ratio of assumed unit inventory cost to the benchmark 3 yuan/pc.year).

Table 6 suggests that when $h_0$ increases from 67% to 333%, the reductions of supply chain costs incurred in LIPWDS model decrease from 7.57% to 1.26%. The computational results in Section 3.1.7 indicates the superiority of centralized inventory

**Table 5**
Costs under different unit order costs for network PB1.

| $A_0$ | LIPWDS | | | LIPWNDS | | | Reductions |
|---|---|---|---|---|---|---|---|
| | Overall | DC Inventory Cost | Retailer Inventory Cost | Overall | DC Inventory Cost | Retailer Inventory Cost | |
| 6 (60%) | 107.3 | 4.0 | 9.0 | 116.7 | 4.2 | 0.6 | 8.03% |
| 8 (80%) | 108.9 | 4.6 | 10.0 | 117.3 | 4.9 | 0.6 | 7.16% |
| 10 (100%) | 110.3 | 5.1 | 10.9 | 117.8 | 5.4 | 0.6 | 6.38% |
| 12 (120%) | 111.6 | 5.6 | 11.8 | 118.3 | 5.9 | 0.6 | 5.67% |
| 14 (140%) | 112.8 | 6.0 | 12.5 | 118.8 | 6.3 | 0.6 | 5.02% |
| 16 (160%) | 113.9 | 6.4 | 13.3 | 119.2 | 6.8 | 0.6 | 4.41% |
| 18 (180%) | 115.0 | 6.8 | 13.9 | 119.6 | 7.2 | 0.6 | 3.84% |
| 20 (200%) | 115.9 | 7.3 | 13.0 | 120.0 | 7.5 | 0.6 | 3.35% |
| 22 (220%) | 116.9 | 7.7 | 13.6 | 120.3 | 7.9 | 0.6 | 2.88% |
| 24 (240%) | 117.7 | 8.0 | 14.1 | 120.7 | 8.2 | 0.6 | 2.44% |
| 72 (720%) | 124.9 | 9.8 | 0.6 | 124.9 | 9.8 | 0.6 | 0 |

strategy over the decentralized strategy. With the increases of unit inventory cost, the increment of invetory costs incurred in the LIPWDS model is greater than that of LIPWNDS. Consequently, LIPWDS model is more benefical in reducing supply chain costs when the unit inventory cost is relatively small.

(3) Sensitivity Analysis of Freight Loss Rate

This research implements a sensitivity analysis by adopting a variety of $\theta_f$.

Table 7 suggests that when $\theta_f$ increases from 5% to 15%, the reductions in supply chain costs by LIPWDS model increase from 1.32% to 5.5%. LIPWDS model is more superior when the freight loss rate is significant. It is because that under LIPWDS model, the number of transit is small and the incurred freight loss is nonsignificant. Considering that fact that currently the freight loss rate in China is about 30%, it is then of great implications to promote the LIPWDS model here.

*Location-Inventory-Routing model*

*Notations, assumptions, and variables*

To simplify the problem, several assumptions are made in this research:

**Table 6**
Costs under different unit inventory costs for network PB1.

| $h_0$ | LIPWDS | | | LIPWNDS | | | Reductions |
|---|---|---|---|---|---|---|---|
| | Overall | DC Inventory Cost | Retailer Inventory Cost | Overall | DC Inventory Cost | Retailer Inventory Cost | |
| 2(67%) | 107.3 | 4.1 | 9.4 | 116.7 | 4.4 | 0.5 | 7.57% |
| 3(100%) | 110.3 | 5.1 | 10.9 | 117.8 | 5.4 | 0.6 | 6.38% |
| 4(133%) | 112.5 | 5.9 | 12.3 | 118.9 | 6.3 | 0.8 | 5.36% |
| 5(167%) | 114.5 | 6.7 | 13.5 | 119.8 | 7.1 | 0.9 | 4.45% |
| 6(200%) | 116.3 | 7.3 | 14.7 | 120.7 | 7.8 | 1.1 | 3.62% |
| 7(233%) | 118.0 | 8.3 | 14.2 | 121.5 | 8.5 | 1.2 | 2.93% |
| 8(267%) | 119.5 | 8.9 | 15.1 | 122.3 | 6.5 | 1.4 | 2.29% |
| 9(300%) | 121.0 | 9.4 | 16.0 | 122.9 | 6.9 | 1.6 | 1.56% |
| 10(333%) | 121.9 | 7.0 | 4.8 | 123.5 | 7.3 | 1.7 | 1.26% |

**Table 7**
Costs under different loss rate for network PB1.

| $\theta_f$ | LIPWDS | | LIPWNDS | | Reduction |
|---|---|---|---|---|---|
| | Overall Cost | Freight Loss | Overall Cost | Freight Loss | |
| 5% | 105.8 | 11.8 | 107.2 | 13.2 | 1.32% |
| 6% | 109.2 | 14.4 | 110.9 | 16.1 | 1.54% |
| 7% | 112.6 | 17.1 | 114.6 | 19.1 | 1.76% |
| 8% | 116.2 | 19.9 | 118.5 | 22.2 | 1.97% |
| 9% | 119.8 | 22.7 | 122.5 | 25.4 | 2.18% |
| 10% | 123.6 | 25.6 | 126.6 | 28.7 | 2.38% |
| 11% | 127.4 | 27.0 | 130.9 | 32.1 | 2.66% |
| 12% | 131.2 | 28.5 | 135.3 | 35.6 | 2.98% |
| 13% | 135.0 | 31.3 | 139.8 | 39.3 | 3.39% |
| 14% | 138.1 | 25.2 | 144.4 | 43.1 | 4.37% |
| 15% | 141.2 | 27.4 | 149.2 | 47.0 | 5.50% |

1) There is no limit of production capability of supplier;
2) Stockout cost is neglected here due to the high service level;
3) Each retailer is served by one delivery route only, also served by one DC/supplier only;
4) Retailer's demand can't exceed the capacity of the delivery vehicle on each delivery route;
5) Delivery vehicles are required to return to the origin;
6) No order lead time is considered in this research.

The notations and variables used in this paper are listed in Table 8.

*Cost analysis incurred in supply chain*

In this research, infrastructure construction costs, transportation costs, inventory costs, and freight losses are considered, with each explained in details below.

(1) Infrastructure construction cost

Construction costs of supplier and DC are:

$$FC = \sum_{b \in B} p_b w_b + \sum_{m \in M} \sum_{b \in B} q_{mb} y_{mb} \tag{50}$$

(2) Transportation cost

Transportation costs are composed of main-line transportation service costs and last-mile delivery costs. Main-line transportation service is applied between supplier and DC while last-mile delivery is implemented between DC and retailer or between supplier and retailer.

**Table 8**
Notations and variables used.

| | | |
|---|---|---|
| ***Set*** | | |
| N | Retailers | |
| M | Distribution centers (DCs) | |
| B | Suppliers | |
| V | Delivery vehicles | |
| S | Union of retailers and DCs | |
| R | Union of retailers and suppliers | |
| U | Union of supplier, retailers and DCs | |
| $L_{mt}$ | Delivery routes of $DC_m$ in day t | |
| $L_{nt}$ | Delivery routes of retailer n in day t | |
| $\Psi_i$ | Journey time to the retailer n | |
| ***Parameters*** | | |
| T | Planning period | day |
| $p_b$ | Construction fee of supplier k within the planning period,k ∈ B | yuan |
| $q_{mb}$ | Construction fee of $DC_m$ served by retailer b within the planning period, m ∈ M, b ∈ B | yuan |
| $c_1$ | Delivery costs per km | yuan/km |
| $c_0$ | Delivery costs per km from supplier to DC | yuan/km |
| $d_{pq}$ | Distance between node p and node q | km |
| $\mu_n$ | Demand of retailer n per day | pcs/day |
| $\Theta$ | Freight loss rate of perishable products from inventory | |
| $h_n$ | Storage costs per pc of retailern ,n ∈ N | yuan/(pc·day) |
| $H_m$ | Storage costs per pc of$DC_m$ ,m ∈ M | yuan/(pc·day) |
| $T_n$ | Delivery cycle time of retailern, n ∈ N | |
| $T_m$ | Order cycle time of$DC_m$ , m ∈ M | |
| $I_n(t)$ | Inventory level of retailer n in day t,n ∈ N, t ∈ $T_i$ | pcs |
| $I_m(t)$ | Inventory level of $DC_m$ in day t,m ∈ M, t ∈ $T_j$ | pcs |
| $r_n(t)$ | Delivery quantity of retailer n in day t,n ∈ N, t ∈ $T_i$ | pcs |
| $\Lambda_m^t$ | Delivery quantity of $DC_j$ in day t,m ∈ M, t ∈ $T_j$ | pcs |
| $ET_n$ | Earliest arrival time at retailer n, n ∈ N | |
| $LT_n$ | Latest arrival time at retailer n, n ∈ N | |
| $ST_n^t$ | Arrival time at retailer n in day t,n ∈ N, t ∈ $T_i$ | |
| $P_L$ | Penalty costs when the vehicle arrives later than $LT_n$ | yuan/hr |
| $P_D$ | Freight loss cost | yuan/pc |
| VC | Capacity limit of delivery vehicle | |
| ***Decision Variables*** | | |
| $w_b$ | 1 if supplier b is selected; 0 otherwise.b ∈ B. | |
| $y_{mb}$ | 1 if $DC_m$ served by supplier b is selected; 0 otherwise. m∈ M, b ∈ B. | |
| $x_{nm}$ | 1 if retailer n is served by$DC_m$; 0 otherwise.n ∈ N, m ∈ M. | |
| $x_{pqv}^{l_{mt}}$ | 1 if vehicle v starts from $DC_m$ and visits node p and q on route $l_{mt}$ in sequence in dayt; 0 otherwise.p ∈ S, q ∈ R, v ∈ V, $l_{mt} \in L_{mt}$. | |
| $z_{nb}$ | 1 if retailer n is served by supplierb; 0 otherwise.n ∈ N, b ∈ B. | |
| $z_{pqv}^{l_{bt}}$ | 1, if vehicle v starts from supplier b and visits node p and q on route $l_{bt}$ in sequence in day t; 0 otherwise.p ∈ S, q ∈ R, v ∈ V, $l_{bt} \in L_{bt}$. | |

$$TC = \sum_{n\in N}\sum_{m\in M}\sum_{b\in B} c_0 d_{mb} y_{mb} + \sum_{m\in M}\sum_{p\in S}\sum_{q\in S}\sum_{v\in V}\sum_{t\in T} c_1 d_{pq} x_{pqv}^{l_{mt}} + \sum_{b\in B}\sum_{p\in R}\sum_{q\in R}\sum_{v\in V}\sum_{t\in T} c_1 d_{pq} z_{pqv}^{l_{bt}} \tag{51}$$

(3) Inventory cost

In this research, ordering costs are included in the inventory costs. Fixed order cycle time and time-varying freight loss is considered here. For retailer n, assume that delivery cycle is$T_i$. Replenishment with quantity of $Q_i$ is made at the beginning of delivery cycle. Inventory level is 0 at the end of cycle. Consequently, it is derived that:

$$\frac{dI_n(t)}{dt} = -\theta I_n(t) - \mu_n \tag{52}$$

with boundary conditions:

$$I_n(0) = Q_n, I_n(T_n) = 0 \tag{53, 54}$$

By solving the differential equation, it is derived that the inventory level of retailer n in day t with the cycle $T_n$ is

$$I_n(t) = \frac{\mu_n}{\theta}\left(e^{\theta(T_n - t)} - 1\right) \tag{55}$$

Let t = 0, Replenishment of retailer n at the beginning of cycle is:

$$I_n = I_n(0) = \frac{\mu_n}{\theta}\left(e^{\theta T_n} - 1\right) \tag{56}$$

Hence the inventory cost of retailer n is:

$$AC_n = \sum_{b\in B} z_{nb}\frac{h_n}{T_n}\int_0^{T_n} I_n(t)dt + \sum_{m\in M} x_{nm}\frac{h_n}{T_n}\int_0^{T_n} I_n(t)dt = \sum_{b\in B} z_{nb}\frac{h_n\mu_n}{T_n\theta^2}\left(e^{\theta T_n} - \theta T_n - 1\right) + \sum_{m\in M} x_{nm}\frac{h_n\mu_n}{T_n\theta^2}\left(e^{\theta T_n} - \theta T_n - 1\right) \tag{57}$$

By applying McLaughlin's expansion$e^{\theta T_n} = 1 + \theta T_n + \frac{1}{2}(\theta T_n)^2$, and $\sum_{m\in M} x_{nm} + \sum_{b\in B} z_{nb} = 1$ stands since the retailer is served by single supplier or DC, it can be derived that $\sum_{m\in M} x_{nm} + \sum_{b\in B} z_{nb} = 1$. Consequently, the running stock cost of retailer i is

$$AC_n = \frac{1}{2}h_n\mu_n T_n \sum_{b\in B} z_{nb} + \frac{1}{2}h_n\mu_n T_n \sum_{m\in M} x_{nm} = \frac{1}{2}h_n\mu_n T_n \tag{58}$$

When the retailer n is served by supplier, the ordering cost is

$$OC_n = \sum_{b\in B} A_n \frac{T}{T_n} z_{nb} \tag{59}$$

It is assumed that for $DC_m$, with the order cycle$T_m$, its inventory level at the beginning of cycle is $I_{m0}$ and $I_{m1}$ at the end of cycle. The delivery quantity in day t is$\Lambda_m^t = \sum_n x_{nm} r_n(t)$. So the inventory in the $DC_m$ satisfies:

$$\frac{dI_m(t)}{dt} + \theta I_m(t) = 0 \quad 0 \leqslant t \leqslant 1 \tag{60}$$

with boundary conditions:

It is derived that the inventory level of $DC_m$ is:

$$Q_m(t) = I_{j0}e^{-\theta t} \quad 0 \leqslant t \leqslant 1 \tag{61}$$

The inventory level of $DC_m$ within the delivery cycle satisfies:

$$I_{m0}^t = I_{m0}^{t-1}\cdot e^{-\theta} - \sum_{n\in N} r_n(t)x_{nm} \quad 0 \leqslant t \leqslant T_m \tag{62}$$

It can be derived that the holding inventory cost of $DC_m$ is:

$$\frac{h_m\int_0^{T_m} I_m(t)dt}{T_m} = \frac{h_m\int_0^{T_m} I_{j0}^t e^{-\theta t}dt}{T_m} = \frac{h_m(1-e^{-\theta})\sum_{t\in T_m} I_{m0}^t}{\theta T_m} \tag{63}$$

Ordering cost of $DC_m$ is:

$$OC_m = \sum_{b\in B} A_m \frac{T}{T_m} y_{mb} \tag{64}$$

Consequently, the overall inventory costs include the retailer's inventory costs, retailer's ordering costs, DC's inventory costs and DC's ordering costs:

$$IC = \sum_{n\in N}\frac{1}{2}h_n\mu_n T_n + \sum_{n\in N}\sum_{b\in B}A_n\frac{T}{T_n}z_{nb} + \sum_{m\in M}\frac{h_m(1-e^{-\theta})\sum_{t\in T_m}I_{m0}^t}{\theta T_m} + \sum_{m\in M}\sum_{b\in B}A_n\frac{T}{T_n}y_{mb} \tag{65}$$

(4) Freight loss costs

The freight loss costs of retailer n are

$$\begin{aligned} &P_D T(I_n(0) - \mu_n T_n\sum_{b\in B}z_{nb} - \mu_n T_n\sum_{b\in B}x_{nm}) = \\ &P_D T\left(\tfrac{\mu_n}{\theta}(e^{\theta T_n}-1) - \mu_n T_n\right) = \\ &\tfrac{P_D T\left[\frac{\mu_n}{\theta}(e^{\theta T_n}-1)-\mu_n T_n\right]}{T_n} \end{aligned} \tag{66}$$

Similarly, the freight loss costs of $DC_m$ can be derived.

$$\frac{TP_D(1-e^{-\theta})\sum_{t\in T_m}I_{m0}^t}{T_m} \tag{67}$$

Consequently, the overall freight loss costs are:

$$C = \sum_{n\in N}\frac{P_D T\left[\frac{\mu_n}{\theta}(e^{\theta T_n}-1)-\mu_n T_n\right]}{T_n} + \sum_{m\in M}\frac{TP_D(1-e^{-\theta})\sum_{t\in T_m}I_{m0}^t}{T_m} \tag{68}$$

By applying McLaughlin's expansion $e^{\theta T_n} = 1+\theta T_n + \frac{1}{2}(\theta T_n)^2$, then

$$MC = \sum_{n\in N}\frac{P_D T\theta T_n\mu_n}{2} + \sum_{m\in M}\frac{TP_D(1-e^{-\theta})\sum_{t\in T_m}I_{m0}^t}{T_m} \tag{69}$$

(5) Overall costs

$$\begin{aligned} STC = FC + TC + IC + MC = &\sum_{b\in B}p_b w_b + \sum_{m\in M}\sum_{b\in B}q_{mb}y_{mb} + \\ &\sum_{n\in N}\sum_{m\in M}\sum_{b\in B}c_0 d_{mb}y_{mb} + \sum_{m\in M}\sum_{p\in S}\sum_{q\in S}\sum_{v\in V}\sum_{t\in T}c_1 d_{pq}x_{pqv}^{l_{mt}} + \\ &\sum_{k\in B}\sum_{p\in R}\sum_{q\in R}\sum_{v\in V}\sum_{t\in T}c_1 d_{pq}z_{pqv}^{l_{bt}} + \sum_{n\in N}\tfrac{1}{2}h_n\mu_n T_n + \\ &\sum_{n\in N}\sum_{b\in B}A_n\tfrac{T}{T_n}z_{nb} + \sum_{m\in M}\tfrac{h_m\left(1-e^{-\theta}\right)\sum_{t\in T_m}I_{m0}^t}{\theta T_m} + \sum_{n\in N}\sum_{\psi_n^t\in\Psi_n}\left[P_E\cdot\max\{ET_n-\psi_n^t,0\} + \left(P_L+P_o\cdot\mu_n\right)\cdot\max\{\psi_n^t-LT_n,0\}\right] \\ &\sum_{n\in N}\tfrac{P_D T\theta T_n\mu_n}{2} + \sum_{m\in M}\tfrac{TP_D\left(1-e^{-\theta}\right)\sum_{t\in T_m}I_{m0}^t}{T_m} \end{aligned} \tag{70}$$

*A mixed integer linear programming model for LIRWDS*

$$\begin{aligned} Min\sum_{b\in B}p_b w_b + &\sum_{m\in M}\sum_{b\in B}q_{mb}y_{mb} + \sum_{m\in M}\sum_{b\in B}c_0 d_{mb}y_{mb} + \\ &\sum_{m\in M}\sum_{p\in S}\sum_{q\in S}\sum_{v\in V}\sum_{t\in T}c_1 d_{pq}x_{pqv}^{l_{mt}} + \\ &\sum_{b\in B}\sum_{p\in R}\sum_{q\in R}\sum_{v\in V}\sum_{t\in T}c_1 d_{pq}z_{pqv}^{l_{bt}} + \sum_{n\in N}\tfrac{1}{2}h_n\mu_n T_n + \\ &\sum_{n\in N}\sum_{b\in B}A_n\tfrac{T}{T_n}z_{nb} + \sum_{m\in M}\sum_{b\in B}A_m\tfrac{T}{T_m}y_{mb} + \\ \sum_{m\in M}\tfrac{h_m\left(1-e^{-\theta}\right)\sum_{t\in T_m}I_{m0}^t}{\theta T_m} + \sum_{n\in N}\sum_{\psi_n^t\in\Psi_n}&\left[P_E\cdot\max\{ET_n-\psi_n^t,0\} + \left(P_L+P_o\cdot\mu_n\right)\cdot\max\{\psi_n^t-LT_n,0\}\right] + \sum_{n\in N}\tfrac{P_D T\theta T_n\mu_n}{2} + \sum_{m\in M}\tfrac{TP_D\left(1-e^{-\theta}\right)\sum_{t\in T_m}I_{m0}^t}{T_m} \end{aligned} \tag{71}$$

S.T.

$$\sum_{m\in M}x_{nm} + \sum_{b\in B}z_{nb} = 1 \quad \forall n\in N \tag{72}$$

$$y_{mb} \leqslant w_b \quad \forall m\in M, b\in B \tag{73}$$

$$z_{nb} \leqslant w_b \quad \forall n\in N, b\in B \tag{74}$$

$$\sum_b y_{mb} \leqslant 1 \quad \forall m\in M \tag{75}$$

$$x_{nm} \leqslant \sum_{b\in B} y_{mb} \quad \forall n \in N, b \in B \tag{76}$$

$$\sum_{q\in S} x_{pqv}^{l_{mt}} = \sum_{q\in S} x_{qp'v}^{l_{mt}} \quad \forall p \in S, p' \in S, v \in V, l_{mt} \in L_{mt} \tag{77}$$

$$\sum_{q\in R} z_{pqv}^{l_{bt}} = \sum_{q\in R} z_{qp'v}^{l_{bt}} \quad \forall p \in S, p' \in R, v \in V, l_{bt} \in L_{bt} \tag{78}$$

$$\sum_{n\in S}\sum_{m\in M}\sum_{v\in V} x_{nmv}^{l_{mt}} \leqslant 1 \quad \forall, l_{mt} \in L_{mt} \tag{79}$$

$$\sum_{n\in S}\sum_{b\in B}\sum_{v\in V} z_{nkv}^{l_{bt}} \leqslant 1 \quad \forall, l_{bt} \in L_{bt} \tag{80}$$

$$x_{nqv}^{l_{mt}} \leqslant x_{nm} \quad \forall n \in N,\ m \in M,\ q \in S,\ v \in V,\ l_{mt} \in L_{mt} \tag{81}$$

$$z_{nqv}^{l_{bt}} \leqslant z_{nb} \quad \forall n \in N,\ b \in B,\ q \in R,\ v \in V,\ l_{bt} \in L_{bt} \tag{82}$$

$$\sum_{p\in S} x_{pqv}^{l_{mt}} \Delta r_n(t) \Delta x_{nm} \leqslant VC \quad \forall m \in M, q \in S, v \in V, t \in T,\ l_{mt} \in L_{mt} \tag{83}$$

$$\sum_{p\in R} z_{pqv}^{l_{bt}} \Delta r_n(t) \Delta z_{nb} \leqslant VC \quad \forall b \in B, q \in R, v \in V, t \in T,\ l_{bt} \in L_{bt} \tag{84}$$

$$I_{m0}^{t} = I_{m0}^{t-1}\Delta e^{-\theta} - \sum_{n\in N} r_n(t)\Delta x_{nm} \quad \forall m \in M, t \in T \tag{85}$$

$$w_b, x_{nm}, y_{mb}, z_{nb}, x_{nmv}^{l_{mt}}, z_{nbv}^{l_{bt}} \in \{0,1\} \quad \forall n \in N, m \in M, b \in B,\ v \in V,\ l_{mt} \in L_{mt},\ l_{bt} \in L_{bt} \tag{86}$$

Eq. (71) is the model objective. Eq. (72) ensures that each retailer is served by one DC or supplier only. Eq. (73) promises that each DC is served from the candidate suppliers only. Eq. (74) promises that each retailer is served from the candidate suppliers only. Eq. (75) ensures that each DC is served by one supplier at most. Eq. (76) ensures that each retailer is served by candidate DCs only. Eq. (77) and (78) keep the continuity of the delivery vehicles, i.e., vehicle can depart only from the node it arrives. Eq. (79) and (80) ensure that there is one vehicle on each route at most. Eq. (81) and (82) promise that the vehicles are used for delivery only if which supplier or DC is identified to serve the retailer. Eq. (83) and (84) ensure that no overload of the vehicle is allowed on each route. Eq. (85) ensures the balance of inventory of DC. Eq. (86) describes 0–1 variables in the model.

*A mixed integer linear programming model for LIRPNDS*

$$\begin{aligned} Min & \sum_{b\in B} p_b w_b + \sum_{m\in M}\sum_{b\in B} q_{mb} y_{mb} + \sum_{m\in M}\sum_{b\in B} c_0 d_{mb} y_{mb} \\ & + \sum_{m\in M}\sum_{p\in S}\sum_{q\in S}\sum_{v\in V}\sum_{t\in T} c_1 d_{pq} x_{pqv}^{l_{mt}} + \sum_{n\in N} \tfrac{1}{2} h_n \mu_n T_n + \sum_{m\in M}\sum_{b\in B} A_m \tfrac{T}{T_m} y_{mb} \\ & + \sum_{m\in M} \frac{h_m\left(1-e^{-\theta}\right)\sum_{t\in T_m} I_{m0}^{t}}{\theta T_m} + \sum_{n\in N}\sum_{\psi_n^t\in\Psi_n} \left[P_E \cdot \max\left\{ET_n - \psi_n^t, 0\right\} + \left(P_L + P_o \cdot \mu_n\right) \cdot \max\left\{\psi_n^t - LT_n, 0\right\}\right] \\ & + \sum_{n\in N} \frac{P_D T \theta T_n \mu_n}{2} + \sum_{m\in M} \frac{TP_D\left(1-e^{-\theta}\right)\sum_{t\in T_m} I_{m0}^{t}}{T_m} \end{aligned} \tag{87}$$

S.T.

$$\sum_{m\in M} x_{nm} = 1 \quad \forall n \in N \tag{88}$$

$$y_{mb} \leqslant b \quad \forall m \in M, b \in B \tag{89}$$

$$\sum_{b} y_{mb} \leqslant 1 \quad \forall m \in M \tag{90}$$

$$x_{nm} \leqslant \sum_{b\in B} y_{mb} \quad \forall n \in N, b \in B \tag{91}$$

$$\sum_{q\in S} x_{pqv}^{l_{jt}} = \sum_{q\in S} x_{qp,v}^{l_{jt}} \quad \forall p \in S, p, \in S, v \in V, l_{jt} \in L_{jt} \tag{92}$$

$$\sum_{n\in N}\sum_{m\in M}\sum_{v\in V} x_{nmv}^{l_{mt}} \leqslant 1 \quad \forall, l_{mt} \in L_{mt} \tag{93}$$

$$\sum_{n\in N} x_{nqv}^{l_{mt}} \Delta r_n(t) \Delta x_{nm} \leqslant VC_v \quad \forall m \in M, q \in S, v \in V, t \in T, \ l_{mt} \in L_{mt} \tag{94}$$

$$I_{m0}^{t} = I_{m0}^{t-1} \Delta e^{-\theta} - \sum_{n\in N} r_n(t) \Delta x_{nm} \quad \forall m \in M, t \in T \tag{95}$$

$$w_b, x_{nm}, y_{mb}, x_{nmv}^{l_{mt}} \in \{0,1\} \quad \forall n \in N, m \in M, b \in B, \quad v \in V, \ l_{mt} \in L_{mt} \tag{96}$$

Eq. (87) is the objective function. Eq. (88) ensures that each retailer is served by one DC only. Eq. (89) ensures that each DC is served by one candidate supplier only. Eq. (90) ensures that each DC is served by one supplier at most. Eq. (91) ensures that each retailer is served by one candidate DC only. Eq. (92) keeps the continuity of the delivery vehicles. Eq. (93) ensures that delivery vehicles can be assigned only if the DC is identified to serve the retailer. Eq. (94) ensures that no overload of the vehicle is allowed on each route. Eq. (95) ensures the balance of inventory of DC. Eq. (96) describes 0–1 variables in the model.

*Solution algorithm*

Since the proposed model is a NP-hard problem, no exact solutions can be derived. Here we propose a two-stage heuristics based on Simulated Annealing algorithm. SA algorithm was initially presented by Kirkpatrick (Kirkpatrick, 1984) which is one of the most popular and effective meta-heuristic algorithms (Kirkpatrick, 1984). The algorithm adopts a stochastic method to fulfill the search by keeping moving to neighboring state even if a worse solution is generated by the move. This outstanding feature is helpful to avoid local optimum traps.

***Step 1***: Initialization;
***Step 2***: Applying the Simulated Annealing algorithm to obtain the solutions of location-allocation problem. Then identify the initial inventory decision and quantify the overall supply chain costs.
***Step 3***: Set the maximum external iteration numbern; Set termination number m when the optimal solution is stable; Set the initial temperature $T_0$ and final temperature $T_1$, temperature decreasing ratio a, $T = T_1{}^{*}a$; .
***Step 4***: Update the retailer allocation using inner neighbourhood operation; Re-optimize the routing using Simulated Annealing algorithm; Update the solution of location-inventory-routing problem;.
***Step 5***: Back to Step 4 when $T > T_0$ and the optimal solution remains stable for less than m iterations;.
***Step 6***: Update the set of candidate DCs/suppliers to open, and some existing DCs/suppliers to shut down. Re-allocate the retailers to the nearest DC/supplier and obtain the new solution;
***Step 7***: Back to Step 4 when the optimal solution remains stable for less than m iterations and external iteration number is less thann;.
***Step 8***: Output the optimal solution.

Firstly, obtain the initial solution for LIRP. Secondly, adopt simulated annealing algorithm to solve location-allocation problem to obtain the strategy for location problem and allocation problem. Take derivation to obtain the optimal ordering cycle. Thirdly, utilize the revised simulated annealing algorithm to obtain the routing strategy for this location-allocation decision. Use local search to optimize by iterations. Finally, use global search to optimize.

(1) Solution of Location-Allocation Problem

The location-allocation model can be derived from LIRP model in Eq. (71).

$$Min \sum_{b\in B} p_b w_b + \sum_{m\in M}\sum_{b\in B} q_{mb} y_{mb} + \sum_{m\in M}\sum_{b\in B} c_0 d_{mb} y_{mb} + \sum_{n\in N}\sum_{m\in M} c_1 d_{nm} x_{nm} \sum_{n\in N}\sum_{b\in B} c_1 d_{nb} z_{nb} \tag{97}$$

S. T.

$$\sum_{m\in M} x_{nm} + \sum_{b\in B} z_{nb} = 1 \quad \forall n \in N \tag{98}$$

$$\sum_{b\in B} y_{mb} \leqslant 1 \quad \forall m \in M \tag{99}$$

$$y_{mb} \leqslant w_b \quad \forall m \in M,\ b \in B \tag{100}$$

$$z_{nb} \leqslant w_b \quad \forall n \in N, b \in B \tag{101}$$

$$x_{nm} \leqslant \sum_{b \in B} y_{mb} \quad \forall n \in N, b \in B \tag{102}$$

$$w_b, y_{mb},\ x_{nm}, z_{nb} \in [0,1] \tag{103}$$

(2) Solution of Optimal Order Cycle and Routing

To get the derivation of Eq. (71), it can be found that.

$$T_i = \sqrt{\frac{2A_n T \sum_{b \in B} z_{nb}}{\mu_n (h_n + P_D T \theta)}} \tag{104}$$

If the retailer n i s served by supplier k, the optimal order cycle of retailer n is.

$$T_n = \sqrt{\frac{2A_n T}{\mu_n (h_n + P_D T \theta)}} \tag{105}$$

Based on the optimal order cycle of DC and retailer, it is then feasible to identify the retailers to be delivered to and optimize the delivery routing accordingly. Simulated Annealing algorithm combined with neighbourhood operator is proposed in this research to obtain the optimized results. To improve the searching efficiency, four types of variable neighborhood search operators are proposed (Fig. 6). According to the number of retailers served by the same DC or supplier, combined strategy of neighborhood search operators is adopted. For example, when there are 2 retailers, Swap operator is chosen. When there are 3 and 4 retailers, Relocate and Swap operators are selected. When there are 5–6 retailers, Relocate, Swap and 2-Opt Operators are selected.

*Numerical example*

This paper applies the proposed model and algorithm to brand fresh milk supply chain comprised with a chain supermarket YH in Beijing. The supply chain includes 1 DC, 41 chain supermarkets and 8 brand candidate milk suppliers. YH was established in 2001 and owns 1032 stores in China. Sales of fresh products account for 55% of the whole sales of YH chain stores. YH has launched about 50 partnerships with local fresh product suppliers around China and 76% of fresh products on their shelves are directly delivered from suppliers. Allowing direct shipment from supplier is of increasing interests to many supply chain members.

The proposed SA heuristic algorithm is coded using Matlab R2014a and run on an Intel Core i53230M (2.60 GHz) processor. The CPU time is 848 s for the proposed case study. The model input includes the demand of each YH chain supermarket (from confidential data resource) and other parameters (Table 9).

*LIRP Considering Direct Shipment from Supplier to Retailer (LIRPWDS)*

(1) Location-allocation decision

The optimal location-allocation decision can be seen in Fig. 7. YH chain supermarkets are divided into three groups and these three groups are served by S1, S3 and S6. From the optimal location-allocation result, it can be seen that all YH chain supermarkets opt for ordering the fresh milk from suppliers directly.

(2) Inventory decision

The optimal order cycle is presented in Table 10. It is found that the optimal order cycle varies with daily demand of each chain supermarket.

(3) Routing decision

Based on the optimal order cycle, it is then feasible to decide whether the shipment from suppliers to the YH chain supermarkets is necessary and optimize the shipment route if necessary. The routing results (Fig. 8) suggest that within the optimized order cycle (i.e., 12 days for each supplier), shipments from suppliers will be made in Day 1, 4, 5, 7, 9 and 10. Noted that all the YH chain supermarkets will be visited in Day 1.

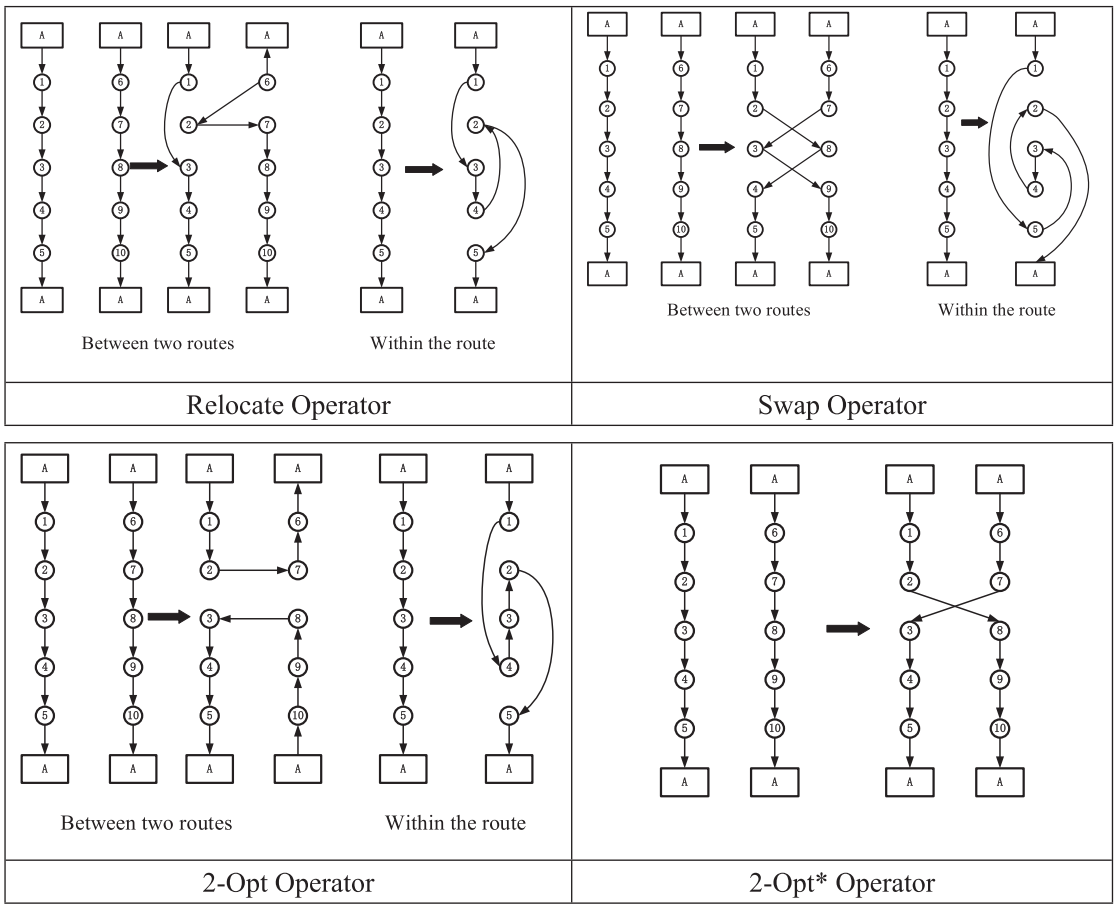



**Fig. 6.** Four neighborhood operators proposed in this research.

**Table 9**
Parameters used in the Case Study.

| Parameters used in the case study<br>Parameters | Definition | Value | Unit |
|---|---|---|---|
| T | Planning period | 360 | day |
| θ | Freight loss rate | 5% | |
| $P_D$ | Freight loss cost per unit | 30 | yuan/pc |
| $P_E$ | Waiting costs when the vehicle arrives earlier than $ET_i$ | 1000 | yuan/hr |
| $P_L$ | Penalty costs when the vehicle arrives later than $LT_i$ | 1000 | yuan/hr |
| $H_j$ | Inventory cost per unit in DC | 20 | yuan/(pc·year) |
| $h_i$ | Inventory cost per unit in retailer | 25 | yuan/(pc·year) |
| $C_0$ | Transportation cost per unit from supplier to DC | 11 | yuan/km |
| $C_1$ | Transportation cost per unit from DC to retailer, or from supplier to retailer | 7 | yuan/km |
| Parameters used in the proposed algorithm | | | |
| n | Max. iteration no. for external loop | 200 | |
| M | Iteration no. when the optimal solution is stable | 30 | |
| $T_0$ | Initial temperature | 10 | |
| $T_1$ | Termination temperature | 10,000 | |
| a | Decreasing ratio of temperature | 0.99 | |

*LIRP without Considering Direct Shipment from Supplier to Retailer (LIRPNDS)*

In this section, it assumed that all YH chain supermarkets receive the goods from YH distribution center, instead of directly ordering from the suppliers. The optimized results suggest that Supplier 2 is selected from the candidate list to serve the DC.

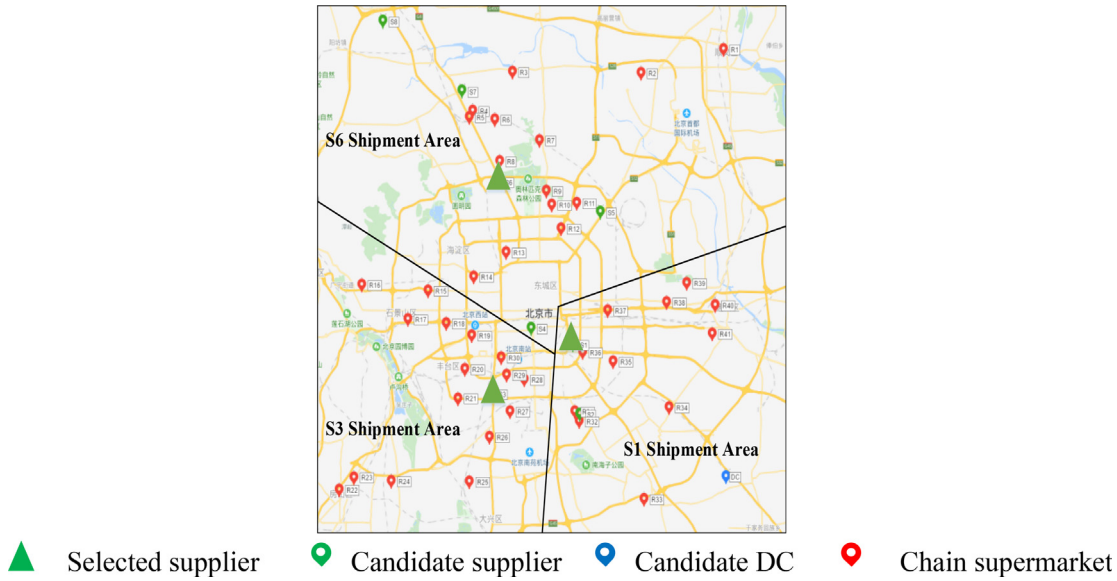



**Fig. 7.** A map showing optimized location-allocation problem in case study.

To further compare the costs incurred in LIRPWDS and LIRPNDS, the costs mentioned here are quantified based on the results obtained in Section 3.2.6. It can be seen from Table 11 that the LIRPWDS are more efficient than LIRPNDS in terms of reducing the overall supply chain costs by 31.1%. Transportation costs account for the largest proportion of overall supply chain cost (42.8% and 43.1% under LIRPNDS and LIRPWDS, respectively). It can then be inferred that reduction in transportation costs is capable of reducing the overall cost significantly. LIRPWDS model is able to save transportation costs by 30.7% compared to LIRPNDS model. It is mainly because that LIRPWDS chain network is more flexible since the supermarkets have the option of ordering the goods from the nearest supplier, while the DC is normally located distant. From the perspective of inventory costs, the LIRPNDS model is more superior because centralized inventory and ordering can utilize the economies of scale of DC. The inventory costs under LIRPWDS model is increased by 99.4% compared to LIRPNDS. Freight loss costs are reduced by LIRPWDS by 69.4% because the goods are not necessarily stored in the DC. In all, LIRPWDS model is proved to be more advantageous in terms of significantly reducing transportation costs and freight losses. With the developments of e-retailers of perishable products, the advantages of LIRPWDS model will be more beneficial, under which the customer satisfaction issues coming directly from the freshness of delivered perishable products. To ensure the freshness of the perishable products, customers can get more fresh products and company can achieve higher level of the customer satisfaction through the implementation of LIRPWDS model.

**Table 10**
Optimal Order Cycle.

| Chain supermarket | R1 | R2 | R3 | R4 | R5 | R6 | R7 | R8 |
|---|---|---|---|---|---|---|---|---|
| Demand | 60 | 58 | 87 | 45 | 87 | 45 | 73 | 52 |
| Order cycle (days) | 4 | 4 | 3 | 3 | 3 | 4 | 3 | 4 |
| Chain supermarket | R9 | R10 | R11 | R12 | R13 | R14 | R15 | R16 |
| Demand | 58 | 69 | 73 | 47 | 87 | 50 | 89 | 54 |
| Order cycle (days) | 3 | 4 | 3 | 4 | 3 | 3 | 3 | 4 |
| Chain supermarket | R17 | R18 | R19 | R20 | R21 | R22 | R23 | R24 |
| Demand | 53 | 51 | 101 | 53 | 53 | 51 | 45 | 49 |
| Order cycle (days) | 3 | 4 | 3 | 3 | 3 | 3 | 3 | 4 |
| Chain supermarket | R25 | R26 | R27 | R28 | R29 | R30 | R31 | R32 |
| Demand | 63 | 93 | 43 | 87 | 43 | 57 | 54 | 53 |
| Order cycle (days) | 3 | 3 | 3 | 3 | 4 | 3 | 4 | 3 |
| Chain supermarket | R33 | R34 | R35 | R36 | R37 | R38 | R39 | R40 |
| Demand | 46 | 59 | 63 | 51 | 64 | 78 | 53 | 58 |
| Order cycle (days) | 4 | 4 | 3 | 3 | 3 | 3 | 4 | 3 |
| Chain supermarket | R41 | S3 | S5 | S6 | | | | |
| Demand | 57 | | | | | | | |
| Order cycle (days) | 4 | 12 | 12 | 12 | | | | |

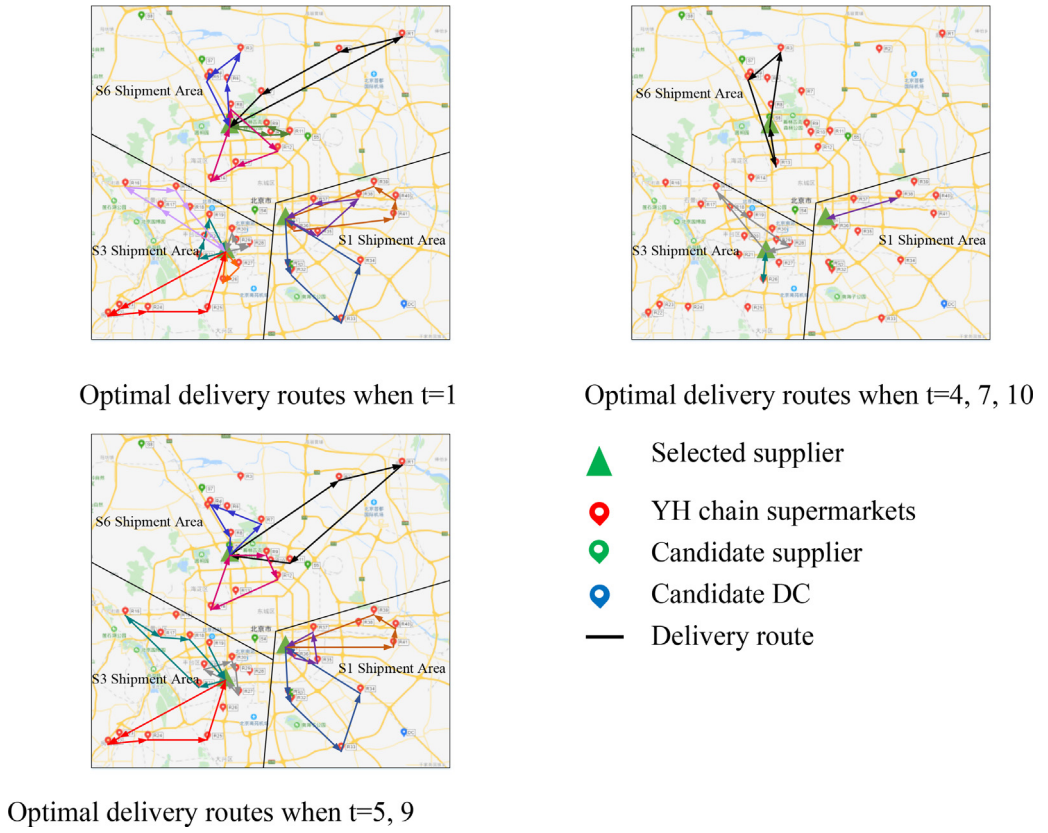



**Fig. 8.** Optimal vehicle routing within an optimized order cycle in LIRPWDS.

**Table 11**
Cost comparison under LIRPWDS and LIRPNDS.

| Cost | Sub-cost | LIRPNDS | LIRPWDS | | Changes |
|---|---|---|---|---|---|
| Transportation Cost | Between supplier and DC | 21240 | NA | | –**30.7%** |
| | Between supplier/DC and retailer | 1343500 | S1 | 264220 | |
| | | | S3 | 321300 | |
| | | | S6 | 353400 | |
| | **SUM** | **1364800** | **945675** | | |
| | Percentage in overall cost | (42.8%) | (43.1%) | | |
| Construction Cost | | **200000** | **300000** | | **50%** |
| | Percentage in overall cost | (6.3%) | (13.7%) | | |
| Inventory Cost | DC ordering cost | 3000 | NA | | –**99.4%** |
| | DC inventory cost | 124210 | NA | | |
| | Retailer inventory costs | 140085 | 140085 | | |
| | Retailer ordering costs | | S1 | 102000 | |
| | | | S3 | 156000 | |
| | | | S6 | 135000 | |
| | **SUM** | **267295** | **533085** | | |
| | Percentage in overall cost | (8.4%) | (24.3%) | | |
| Freight Loss Cost | | **1351800** | **413505** | | |
| | Percentage in overall cost | (42.5%) | (18.9%) | | –**69.4%** |
| Overall Cost | | **3183895** | **2192265** | | –**31.1%** |

## Conclusions

The rapid developments of grocery market, especially the perishable product market in China, has motivated the current research to investigate the simultaneous optimization of several decisions involved in the supply chain for perishable products, including location, inventory and routing strategy. A three-echelon supply chain including suppliers, DCs and retailers

**Table 12**
Main Conclusions.

| | LIPWDS | LIPWNDS | Comparison |
|---|---|---|---|
| **LIP** | ***Transportation*** is the most dominant component, accounting for 51% of overall cost. Inventory shares 18% of overall. | ***Transportation*** is the most dominant component, accounting for 50% of overall cost. Inventory shares 4% of overall. | Allowance of direct shipment reduces 6.38%-16.93% of overall SC cost, depending on the network selected. |
| | Dealer ***inventory*** and ordering cost of retailers account for 89% of overall inventory cost. | Dealer ***inventory*** and ordering cost of DCs account for 87% of overall inventory cost. | LIPWDS model is able to achieve lower single ordering cost, single inventory cost. LIPWDS is more superior when freight loss rate is relatively high. |
| | Optimal ***order quantity*** under centralized inventory strategy is less than that of decentralized strategy. | ***Safety stock*** of DCs might vary significantly, depending on the scale of DC. | LIPWDS model is able to reduce transportation and freight loss costs whereas increase inventory and infrastructure construction costs. |
| **LIRP** | ***Transportation*** is the most dominant component, accounting for 43.1% of overall cost. Inventory shares 24.3% of overall. | ***Transportation*** is the most dominant component, accounting for 42.9% of overall cost. Freight loss cost shares 42.5% of overall. | Allowance of direct shipment reduces 31.1% of overall SC cost. |
| | Routing decision has great impact on whole supply chain performance. | Ordering all the goods by DC is able to minimize the whole SC inventory. | LIRPWDS model reduces the transportation cost by 30.7% and reduces freight losses by 30.7%. 69.4%. |

is structured in this paper, with the impacts of whether allowing direct shipment from supplier to retailer being considered. Two typical decision problems are tackled here: LIP and LIRP. The optimization results in Table 12 suggest that the location, inventory and routing decisions can be optimized simultaneously through the proposed integrated approach. 1) Both LIPWDS and LIRPWDS models are able to reduce the overall supply chain costs, compared with LIPWNDS and LIRPWNDS. 2) For both LIP and LIRP, whether considering the direct shipment from supplier to retailer, transportation cost is the most significant component of the overall supply chain costs, accounting for more than 40% of overall. 3) Both LIPWDS and LIRPWDS models are able to reduce the transportation costs, compared with LIPWNDS and LIRPWNDS. However, the inventory costs are increased by LIPWDS and LIRPWDS.

There are some limits of current research. Firstly, the model objective is to minimize the overall supply chain costs. The inclusion of service level should be considered in the future research. Secondly, dynamic demand of retailers for perishable products should be further investigated in the LIRPWDS model. Thirdly, extension of current three-echelon supply chain network to the four-echelon network should be implemented.

## Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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